

The influence of temporal context on vision over multiple time scales

Kacie Lee, Reuben Rideaux 

School of Psychology, The University of Sydney, Camperdown, Australia • Queensland Brain Institute, The University of Queensland, St Lucia, Australia

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This **fundamental** study shows how past experiences shape perception across short, medium, and long time scales, using a single behavioural paradigm and reanalysed EEG data. It provides **convincing** evidence for two processes across all scales: an attention-dependent mechanism that speeds responses to expected events, and an attention-independent mechanism where expected events are encoded less precisely, consistent with feedforward dampening. The work offers a unifying account of temporal context effects, though stronger brain-behaviour links, integration with serial dependence attraction and repulsion models, and extension to other timescale definitions would further strengthen the contribution.

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Abstract

Past sensory experiences influence perception of the present. Multiple research subfields have emerged to study this phenomenon at different temporal scales. These phenomena fall into three categories: the influence of immediately preceding sensory events (micro), expectations established by short sequences of events (meso), and regularities over long sequences of events (macro). In a single paradigm, we examined the influence of temporal context on perception at each scale. By integrating behavioural and pupillometry recordings with electroencephalographical recordings from a previous study, we identify two distinct mechanisms that operate across all scales. The first is moderated by attention and supports rapid motor responses to expected events. The second operates independently of task-demands and dampens the feedforward neural responses produced by expected events, leading to unexpected events eliciting earlier and more precise neural representations.

Introduction

The capacity to adapt to patterns in the environment supports biological function from sensory processing to motor action. The temporal context in which sensory events are embedded can be leveraged to more effectively process and respond to this information. For example, expert tennis players predict the trajectory of the ball from the movements of their opponent and use this

information to prepare their return (1 [↗](#)). Influential theories of normative brain function such as predictive coding posit that temporal context serves to improve representational fidelity and reduce neurometabolic expenditure (2 [↗](#)–4 [↗](#)).

Humans are sensitive to the temporal context of events across multiple time scales. At the shortest scale, each event influences processing of the next event (**Fig. 1a** [↗](#), micro). For example, stimulus reproductions are biased towards previous stimuli. This phenomenon is referred to as *serial dependency* and has been demonstrated for a range of visual and auditory features (5 [↗](#)–7 [↗](#)). At intermediate scales, short sequences of events form patterns that uniquely influence responses to subsequent events (**Fig. 1a** [↗](#), meso). For instance, events that satisfy regular patterns produce attenuated neural responses (*oddball effect*) (8 [↗](#)) and are responded to more quickly and accurately than events that violate them (*sequential dependencies*) (9 [↗](#), 10 [↗](#)). At longer time scales, the relative frequency of past events can alter how those in the present are processed (**Fig. 1a** [↗](#), macro). These effects are often referred to as *statistical learning* and are thought to reflect adaptation to regularities in the environment (11 [↗](#), 12 [↗](#)).

Incoming sensory information is shaped by its temporal context across time scales that range many orders of magnitude. Previous work has developed a variety of experimental designs to isolate the influence of temporal context at each scale and examine its behavioural and neural consequences. This approach has propagated multiple fields of research and implicitly asserts qualitative differences, however, it limits comparison between temporal context at different scales. It is possible that temporal context serves perception differently at each time scale and is thus associated with unique adaptive influences on sensory processing. For example, perception may be best supported by one mechanism that promotes a set of outcomes from moment to moment and another mechanism that promotes a different set of outcomes in response to trends formed over long periods. Alternatively, a common rule may be applied at all scales of temporal context, signalling a unifying adaptive mechanism. Predictive coding theory offers a normative framework with which to understand the influence of temporal context across scales, but contradictory evidence (3 [↗](#), 13 [↗](#)–15 [↗](#)) (i.e., sharpening or dampening of expected events) from distinct experimental paradigms has challenged this unification.

We designed a novel paradigm to investigate how temporal context shapes perception across multiple scales. We further used electroencephalography (EEG) and pupillometry recordings to characterize the neural mechanisms associated with these perceptual consequences. We identify two distinct mechanisms that operate across all scales. The first is moderated by attention and supports rapid motor responses to expected events. The second is independent of task-demands and dampens the feedforward neural responses to expected events, leading to unexpected events eliciting earlier and more precise neural representations. Together, these adaptive mechanisms explain a raft of temporal sensory phenomena, and provide evidence for a dampening account of predictive coding.

Results

To measure changes in visual perception associated with different scales of temporal context, we tasked participants with indicating the location of serially presented visual stimuli (gaussian blobs randomly positioned at a fixed distance around a central fixation point). To assess response time and task accuracy, participants performed a speeded binary judgement (e.g., left or right of fixation) on each trial. On 10% of trials, participants additionally reproduced the location of the stimulus, providing a measure of recall precision. To test the influence of attention, trials were sorted according to two spatial reference planes, based on the location of the stimulus: task-related and task-unrelated (**Fig. 1b** [↗](#)). The task-related plane corresponded to participants' binary judgement (**Fig 1b** [↗](#), light cyan vertical dashed line) and the task-unrelated plane was orthogonal to this (**Fig 1b** [↗](#), dark cyan horizontal dashed line). For example, if a participant was tasked with

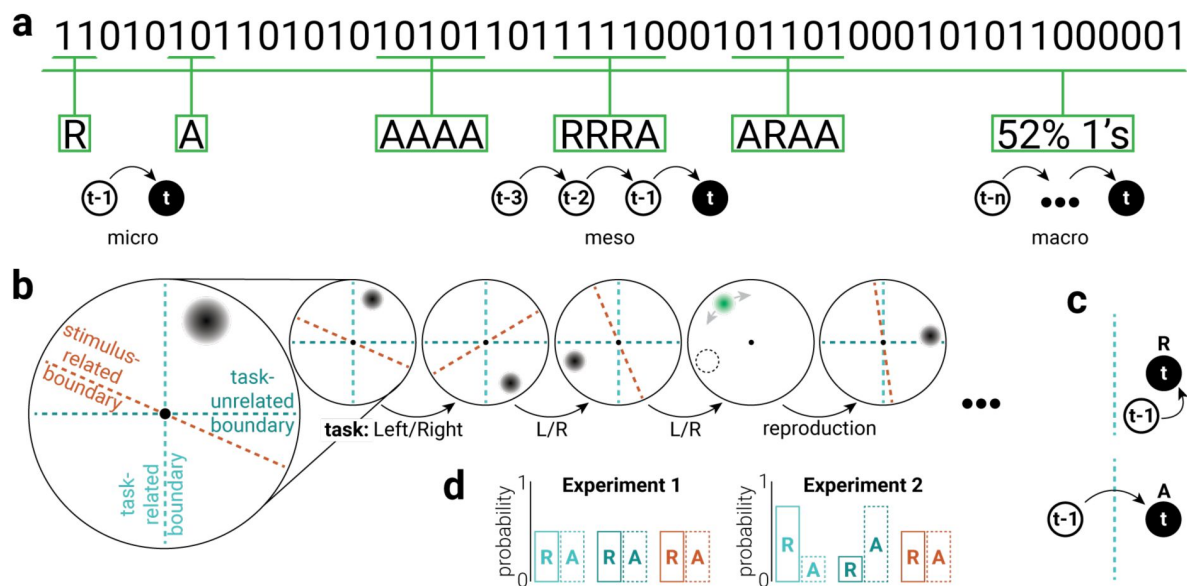


Figure 1.

Multiple scales of temporal context.

a) Illustration of three scales of temporal context in a binary time series. At the micro scale, events can either stay the same over time (repeat; R) or change (alternate; A). At the meso scale, short sequences of events (e.g., 5) form patterns that vary in their regularity (e.g., four repeats of the same event seems more regular than a mixture of repeats and alternations between events). At the macro scale, general trends regarding the relative frequency of different events are formed over longer time periods. **b)** To test the influence of temporal context on visual perception across different scales, participants were instructed to indicate the location of serially presented targets, which were randomly positioned on an imaginary circle centered on fixation. Participants performed a speeded binary judgement (e.g., left/right of fixation) on each trial and additionally reproduced the location of the target on 10% of trials. Trials were categorized as either (**c**, top) repeat (R) or (**c**, bottom) alternate (A) based on the location of the target relative to the previous target, according to three spatial reference planes: task-related (light cyan), task-unrelated (dark cyan), and stimulus-related (orange). **d)** In Experiment 1, the location of targets was uniformly sampled such that repeat and alternate trials were equally likely. In Experiment 2, the probability of repeat and alternate trials was biased across the task-related and unrelated planes.

performing a left-or-right-of-fixation judgement, then their task-related plane was the vertical boundary between the left and right side of fixation, while their task-unrelated plane was the horizontal boundary. The former (left-right) axis is relevant to their task while the latter (top-bottom) axis is orthogonal and task irrelevant. This orthogonality can be leveraged to analyze the same data twice (once according to the task-related plane and again according to the task-unrelated plane) in order to compare performance when the relative location of an event is either task relevant or irrelevant.

Micro-scale temporal context

At the shortest temporal scale, the speed with which events are responded to can be influenced by those that immediately precede them (16). We investigated the influence of the previous stimulus on the current stimulus by comparing responses between trials in which the previous stimulus was presented either on the same side (repeat) or the other side (alternate) of the reference plane (Fig. 1c, 2a). Consistent with previous work (16), we found faster and more accurate responses to repeat stimuli along the task-related plane (speed: $z_{75}=2.33$, $p=.020$, $r=.27$; task accuracy: $z_{72}=2.41$, $p=.016$, $r=.28$; Fig. 2b-c, light cyan). However, in contrast, we found that repeat stimuli were recalled less precisely than alternate stimuli ($z_{72}=3.74$, $p=2.00e^{-4}$, $r=.44$; Fig. 2d, light cyan). For the task-unrelated plane, we found faster responses for repeat stimuli, but no difference in task accuracy, and marginally higher precision for alternate stimuli (speed: $z_{75}=6.66$, $p=2.76e^{-11}$, $r=.77$; accuracy: $z_{72}=0.92$, $p=.357$, $r=.11$; precision: $z_{76}=1.62$, $p=.105$, $r=.19$; Fig. 2b-d, dark cyan).

While the difference in response time for repeat and alternate stimuli were of similar magnitude between task-related and unrelated planes ($z_{71}=1.15$, $p=.249$), the *variance* across participants was considerably higher in the former condition ($F_{1,140}=46.26$, $p=2.76e^{-10}$). The increased variability of response time differences across the task-related plane likely reflects individual differences in attention and prioritization of responding either quickly or accurately. On each trial, the correct response (e.g., left or right) was equally probable. So, to perform the task accurately, participants were motivated to respond without bias, i.e., without being influenced by the previous stimulus. We would expect this to reduce the difference in response time for repeat and alternate stimuli across the task-related plane, but not the task-unrelated plane. However, task-relatedness may amplify the bias towards making faster responses for repeat stimuli, by increasing attention to the identity of stimuli as either repeats or alternations (17). These two opposing forces vary with task engagement and strategy and thus would be expected produce increased variability across the task-related plane. Unlike response time, we would not expect task accuracy to be modulated by task-unrelated micro temporal context, as the tasks were orthogonal. That is, while a repeat along the task-unrelated plane may increase the speed with which an (in/correct) response is made along the task-related plane, it would not be expected to change the type of response (e.g., left or right). Precision differences were similar between task-related and unrelated planes, both in terms of magnitude ($z_{70}=1.77$, $p=.076$) and variability ($F_{1,138}=0.66$, $p=.417$), suggesting that modulation of representational fidelity may be operationalized by a distinct mechanism to that which modifies response speed, and is not moderated by attention.

These results could be interpreted as a general expectation that sensory events, e.g., the image of an object on the retina, are more likely to stay the same than to change (7). Through this lens, these findings would indicate that micro temporal context leads to faster and more accurate responses to expected events, but more precise encoding of unexpected events. As we will show, these two outcomes resonate across all levels of temporal context and point to two distinct mechanisms that unify the influence of past experiences on perception.

To test for neural correlates of micro temporal context, we re-analyzed a previously published EEG dataset in which human participants viewed visual (arc) stimuli presented at random angles around fixation, while monitoring for targets (stimuli presented at a specific location) (17). We used multivariate linear-discriminant analysis to classify the location of stimuli according to task-

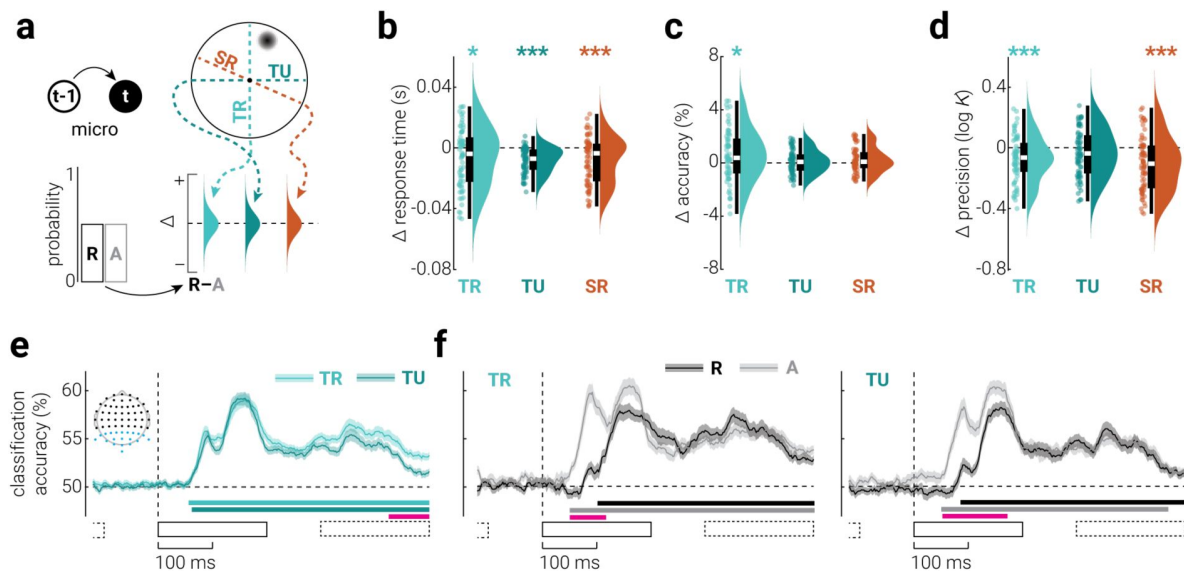


Figure 2.

The influence of micro-scale temporal context on visual processing.

a) Micro temporal context refers to the influence of the last event on the current event. We assessed its influence by comparing task performance for repeat and alternate presentations along task-related (light cyan), task-unrelated (dark cyan), and stimulus-related (orange) reference planes. **b-d)** The difference (repeat – alternate) in **(b)** response time, **(c)** task accuracy, and **(d)** precision for the three reference planes. Asterisks indicate significant differences (* $p < .05$, ** $p < .01$, *** $p < .001$). **e)** Classification accuracy of stimuli presented on different sides of task-related and unrelated planes, from re-analysis of previously published EEG data (17). **f)** Same as **(e)**, but split into repeat and alternate stimuli, along (left) task-related and (right) unrelated planes. Inset in **(e)** shows the EEG sensors included in the analysis (blue dots). Black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). Shaded regions indicate $\pm$ SEM. Horizontal bars indicate cluster corrected periods of significance (cyan and greyscale: above chance classification accuracy, pink: difference).

related ($< \text{or} > |90^\circ|$ from target angle) and unrelated planes (orthogonal to task-related plane), from parietal and occipital sensors (**Supplementary Fig. 1a**). We found that classification accuracy was higher across the task-related plane (**Fig. 2e**).

In separately assessing classification accuracy for repeat and alternate stimuli, we found temporal dependencies that were due to low frequency signals, unrelated to stimulus properties, as evidenced by their presence in results produced from data with shuffled labels (see **Supplementary Fig. 2** for a full description of the temporal dependencies). We removed these general temporal dependencies by subtracting the bias in the shuffled data from the original classification accuracy. For both task-related and unrelated planes, we found that classification accuracy was higher for alternate stimuli from ~ 60 – 160 ms (i.e., the first moments of reliable stimulus decoding) and that alternate stimuli could be reliably classified ~ 20 ms earlier than repeat stimuli (**Fig. 2f**). The difference in classification accuracy between repeat and alternate stimuli was the same for task-related and unrelated conditions. This neural phenomenon may explain the increased behavioural recall precision observed for alternate stimuli. The timing and task-independence of this phenomenon points to a passive, massively parallel, mechanism that operates on feedforward sensory signals, by suppressing responses to repeat stimuli.

Another phenomenon associated with micro temporal context is *serial dependence*, i.e., reproductions of events from memory are biased towards the events that immediately preceded them (5, 7). To assess serial dependence in the behavioural data, we began by comparing behavioural performance across a third, stimulus-related, plane (**Fig 2a**, orange); whereas task planes were constant, the stimulus-related plane was defined by the location of the stimulus on the previous trial, and thus varied from trial to trial. That is, on each trial, the target is considered a repeat if it changes location by $< |90^\circ|$ relative to its location on the previous trial, and an alternate if it changes by $> |90^\circ|$. We found a similar pattern of results as those for task-related and unrelated planes: responses were faster for stimuli that repeated ($z_{77}=3.73$, $p=2.00e^{-4}$, $r=.43$; **Fig. 2b**, orange) but more precise for those that alternated ($z_{76}=4.51$, $p=6.34e^{-6}$, $r=.52$; **Fig. 2d**, orange). The latter result is inconsistent with previous work on serial dependence, which has reported more precise judgements when subsequently presented stimuli are similar (18).

To further investigate the influence of micro temporal context, we computed task performance as a function of the angular distance from the previous stimulus (Δ location). We calculated the bias of reproduction responses as a function of Δ location and found the archetypal pattern of attractive biases associated with serial dependence (**Fig. 3a**). Consistent with our previous binary analysis, and with previous work (19), we also found that responses were faster and more accurate when Δ location was small (**Fig. 3b, c**). We further found that the precision of responses peaked at both large and small Δ locations, with the worst precision for stimuli with around $\pm 60^\circ$ Δ location (**Fig. 3d**). The peak in precision for large Δ locations is consistent with alternate events being encoded more precisely, while the peak for small offsets may be explained the attractive bias towards the previous target.

To further investigate the influence of serial dependence, we applied inverted encoding modelling to the EEG recordings to decode the angular location of stimuli. We found that decoding accuracy sharply increased from ~ 60 ms following stimulus onset (**Fig. 3e**). Note, to remove the influence of general temporal dependencies, we applied a 0.7 Hz high-pass filter to the data, which temporally smeared the stimulus-related information, resulting in above-chance decoding accuracy prior to stimulus presentation (for full details, see **Supplementary Figure 3**). To understand how serial dependence influences the representation of these features, we inspected decoding accuracy as a function of both time and Δ location (**Fig. 3f**). We found that decoding accuracy varied depending not only as a function of time, but also as a function of Δ location. To characterise the latter relationship, we calculated the average decoding accuracy from 50 ms until the end of the epoch (500 ms), as a function of Δ location (**Fig. 3g**). This revealed higher accuracy for targets with larger Δ location. We found a similar pattern of results for decoding

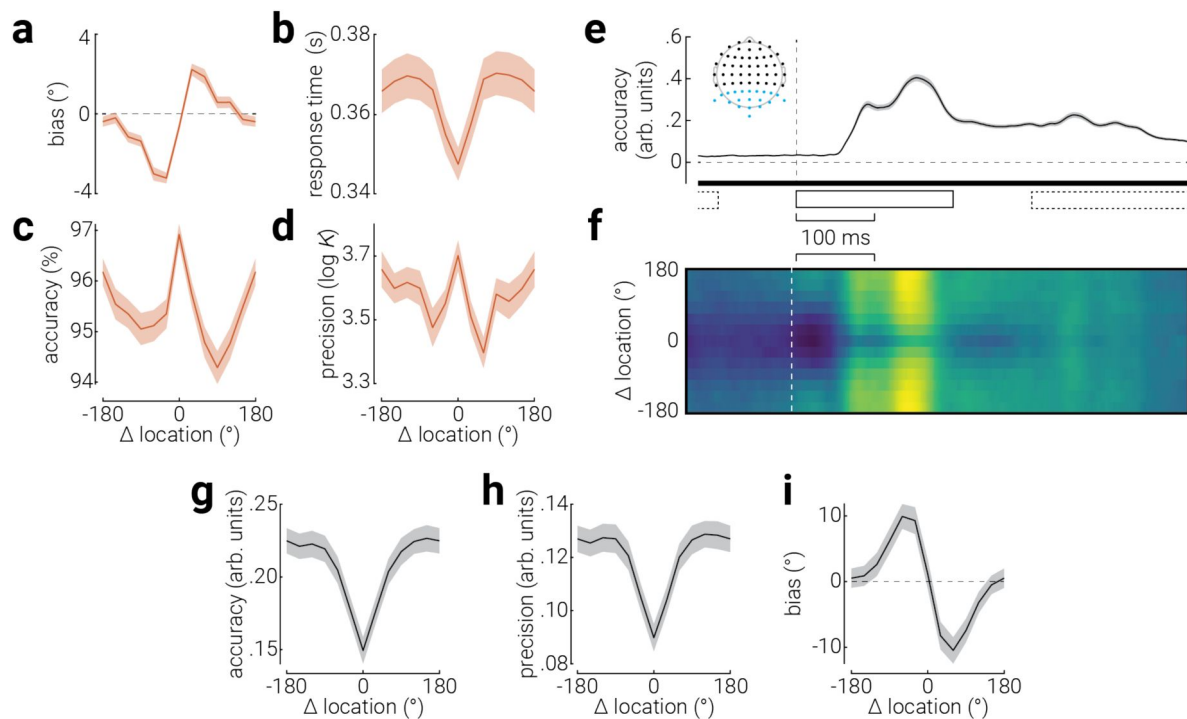


Figure 3.

The influence of serial dependence on visual processing.

Serial dependence is associated with sensory stimuli being reported as more similar to previous stimuli and is typically assessed by measuring the perception of stimuli as a function of their distance to previous stimuli (Δ location). **a**) Location reproduction bias, **b**) binary task response time and **c**) accuracy, and **d**) reproduction precision as a function of distance from the previous stimulus. **e**) Decoding accuracy for stimulus location, from re-analysis of previously published EEG data (17). Inset shows the EEG sensors included in the analysis (blue dots), and black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). **f**) Decoding accuracy for location, as a function of time and Δ location. Bright colours indicate higher decoding accuracy; absolute accuracy values can be inferred from **e**). **g-i**) Average location decoding **(g)** accuracy, **(h)** precision, and **(h)** bias from 50 – 500 ms following stimulus onset. Horizontal bar in **e**) indicates cluster corrected periods of significance; note, all time points were significantly above chance due to temporal smear introduced by high-pass filtering (see **Supplementary Figure 3** for full details). Note, the temporal abscissa is aligned across **e** & **f**). Shaded regions indicate $\pm$ SEM.

precision (**Fig. 3h**). These results are consistent with the micro temporal context (behavioural) results, showing that targets that alternated were recalled more precisely. Lastly, we calculated the decoding bias as a function of Δ location and found a clear repulsive bias away from the previous stimulus (**Fig. 3i**). While this result is inconsistent with the attractive behavioural bias, it is consistent with recent studies of serial dependence suggesting an initial pattern of repulsion followed by an attractive bias during the response period (20–22).

Meso-scale temporal context

The influence of recent events can accrue over time, producing unique changes in behavioural responses associated with short sequences of stimuli (**Fig. 4a**). Higher-order sequential dependencies are an example of how stimuli (at least) as far back as five events in the past can shape the speed and task accuracy of responses to the current stimulus (9, 10); however, note that these effects have been observed for more than five events (23). For five binary events (e.g., left or right), there are 16 possible sequences of repeat and alternate stimuli. Typically, sequential dependencies are assessed by computing the average response time and/or task accuracy for stimuli, binned according to which of these 16 sequences reflects their recent history. The differences observed between sequences have been interpreted as indicative of the expectation of the current stimulus, based on previous events (9).

We tested the influence of higher-order sequential dependencies on response time, task accuracy, and recall precision, across task-related and unrelated reference planes. However, we first isolated the influence of meso temporal context from that of micro context (i.e., difference in performance between repeat and alternate stimuli), by binning responses into eight sequences formed by combining the symmetric partners in the 16-sequence array (e.g., RRRR with AAAA, RRRR with AAAR, etc.). Thus, the frequency of repeat and alternate stimuli was equal across the eight sequences and any differences observed in performance are unlikely to be due to differential responses to these events. We found that responses were faster for sequences with expected targets across both reference planes (task-related: $\chi^2_{7,469}=375.86$, $p=3.56e^{-77}$, $W=.84$; task-unrelated: $\chi^2_{7,483}=110.94$, $p=5.85e^{-21}$, $W=.26$; **Fig. 4b**). Responses were also more accurate for expected targets across the task-related plane ($\chi^2_{7,399}=208.37$, $p=2.03e^{-41}$, $W=.44$; **Fig. 4c**). By contrast, we found no such clear relationship between precision and expectancy (both $p>.05$; **Fig. 4d**).

We found that micro temporal context influenced recall precision, whereas we did not detect significant modulation of precision for meso context. This may be because meso temporal effects reflect motor priming and do not precipitate visual predictions. When predictions are violated, the pupils dilate in what has been interpreted as an arousal response (24, 25). We found that participants' pupils dilated more in response to unexpected sequences (**Fig. 4d**), indicating that meso temporal context establishes sensory predictions. Another possible explanation why we did not detect an effect on recall precision is that we lacked statistical power. The micro analyses split data between two conditions (repeat and alternate), whereas in the meso analysis, data is split between eight (sequences). Indeed, in our second experiment, we reproduced the response time and task accuracy effects of meso temporal context and further found that unexpected sequences were recalled more precisely (**Supplementary Fig. 4**).

To further test whether meso temporal context effects were qualitatively different from those at the micro scale, we separated the EEG classification accuracies (**Fig. 2e**) into the eight sequence categories and corrected for shuffled accuracy (**Supplementary Fig. 5**). We found a strikingly similar pattern of results to those observed at the micro scale: there was increased classification accuracy for unexpected sequences from ~60-100 ms along both task-related and unrelated reference planes (**Fig. 4h, g**). This finding provides further evidence of a passive mechanism that suppresses feedforward sensory activity elicited by expected stimuli and leads to more

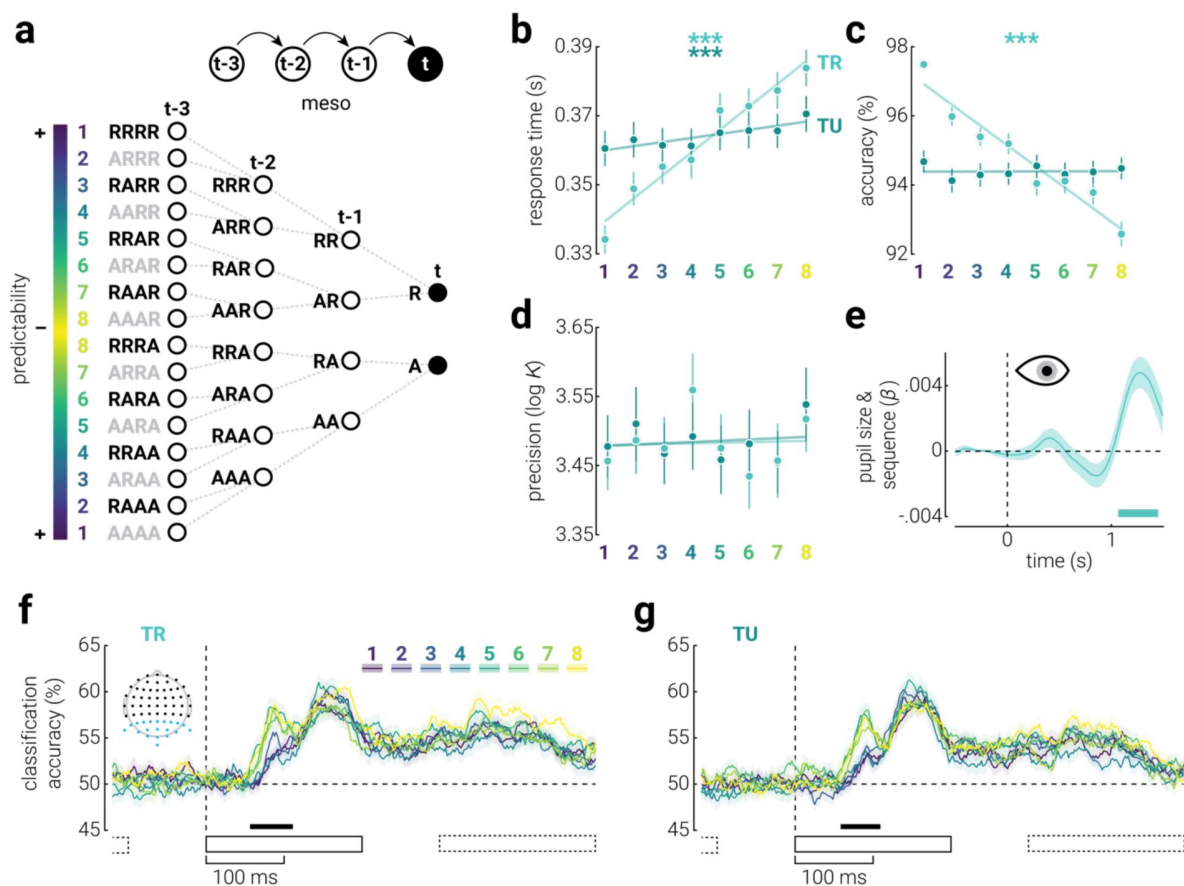


Figure 4.

The influence of meso-scale temporal context on visual processing.

a) Illustration of all 16 possible sequences of repeat and alternate events for a series of five binary events. Sequences are arranged symmetrically such that those for which the final event is thought to be most expected are at the top and bottom and those for which the final event is least expected are in the middle. To control for differences between repeat and alternate events, we combined symmetric pairs, resulting in a total of eight sequences. **b)** Response time, **(c)** task accuracy, and **(d)** precision as a function of sequence, across task-related and unrelated reference planes. Note, lower numbers on the abscissa are associated with sequences in which the final target stimulus is more expected. Asterisks indicate significant main effects of sequence ($***p < .001$). Error bars indicate $\pm$ SEM; semi-transparent lines indicate linear fits to the data. **e)** The correlation between pupil size and sequence as a function of time. Note, we did not analyze pupillometry data for micro temporal context due the confounding effect of differential foveal luminance between repeat and alternate stimuli. **f)** Classification accuracy of stimuli presented on different sides of task-related planes as a function of time, for each of the eight sequences, from re-analysis of previously published EEG data (17). **g)** Same as **(f)**, but for the task-unrelated plane. Inset in **(f)** shows the EEG sensors included in the analysis (blue dots). Black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). Shaded regions indicate $\pm$ SEM. Horizontal bars indicate cluster corrected periods of significant relationships between **(e)** pupil size or **(f, g)** classification accuracy and sequence order.

precise recall of those that are unexpected. That the effect was reproduced when repeat and alternate stimuli were matched also shows that it cannot be explained by sensory adaptation (26 [↗](#)).

Macro-scale temporal context

Events within micro and meso temporal contexts occur within the limits of short-term memory. Most events that occur further into the past cannot be recalled; yet, they continue to influence perception (11 [↗](#), 12 [↗](#)). To test the influence of macro temporal context on perception, we ran a second experiment in which the probability of stimulus presentation was biased towards repeats or alternates along task-related and unrelated reference planes (**Fig. 1d** [↗](#), Experiment 2). The biases were counterbalanced across participants such that we could compare performance between relatively frequent (expected) and infrequent (unexpected) events, while controlling for differences between repeat and alternate stimuli (**Fig. 5a** [↗](#)). Consistent with micro and meso scale effects, we found that expected events were responded to faster and more accurately, while unexpected events were reproduced more precisely, across both task-related (speed: $z_{78}=7.67$, $p=1.68e^{-14}$, $r=.87$; task accuracy: $z_{73}=7.42$, $p=1.13e^{-13}$, $r=.87$; precision: $z_{79}=3.26$, $p=.001$, $r=.37$) and unrelated planes (speed: $z_{71}=6.69$, $p=2.19e^{-11}$, $r=.79$; task accuracy: $z_{78}=6.76$, $p=1.32e^{-11}$, $r=.77$; precision: $z_{75}=2.72$, $p=.007$, $r=.31$; **Fig. 5b-d** [↗](#)). This effect was larger along the task-related plane for response time and task accuracy, but not precision (speed: $z_{71}=7.24$, $p=4.42e^{-13}$, $r=.86$; task accuracy: $z_{78}=7.27$, $p=3.52e^{-13}$, $r=.82$; precision: $z_{75}=0.93$, $p=.350$, $r=.11$). Pupillometry analysis further revealed that participants' pupils dilated more in response to unexpected stimuli ~ 1 s after stimulus onset (**Fig. 5e** [↗](#)).

In line with previous work, stimuli in our experiment were displayed until participants responded. To test how variable presentation duration influenced the results, we ran a third experiment that was the same as the second, except with a fixed stimulus duration (200 ms). Experiment 3 replicated both the behavioural and pupillometry results from Experiment 2 (**Supplementary Fig. 6** [↗](#)), indicating that the effects observed were not related to variable stimulus duration.

Discussion

Evidence from multiple, separate, fields of research show that humans are sensitive to the temporal context in which sensory stimuli are embedded across a broad range of time scales. Here we used a novel experimental paradigm to characterise and compare how past sensory events occurring at different temporal scales shape perception of the present. By combining behavioural, neuroimaging, and physiological approaches to assess the influence of temporal context at the micro, meso, and macro scale, we identified new properties of these phenomena in isolation and further revealed two distinct adaptive mechanisms that unify them.

Micro-scale temporal context

At the micro scale, stimuli were responded to faster and more accurately when preceded by those that were similar, but recalled more precisely when preceded by those that were different. This may reflect a general expectation that objects in the environment are more likely to stay the same than to change (7 [↗](#)). Under this interpretation, expected events are rapidly responded to while unexpected events are precisely encoded. As will be discussed shortly, this interpretation resonates with the influence of temporal context at longer scales.

Analysis of EEG recordings showed that during the initial stage of stimulus processing in the cortex (60 ms after onset), alternating stimuli were represented both (20 ms) earlier and more robustly than those that repeat. This finding provides a neural correlate of the improved recall

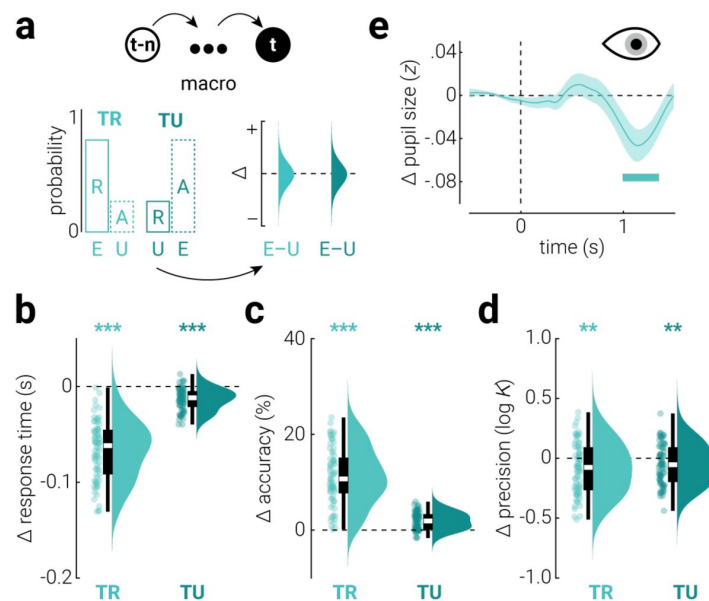


Figure 5.

The influence of macro temporal context on visual processing.

a) Repeat (R) and alternate (A) stimuli were presented with unequal probabilities (counterbalanced across participants) along task-related and unrelated reference planes. The difference in performance between relatively frequent (expected; E) and infrequent (unexpected; U) stimuli was calculated to assess the influence of macro temporal context on visual processing. **b-d)** The difference (expected – unexpected) in **(b)** response time, **(c)** task accuracy, and **(d)** reproduction precision between expected and unexpected stimuli, along task-related (TR) and unrelated (TU) planes. Note that we observed task accuracy differences associated with macro temporal context across the task-unrelated plane. In the micro and meso analyses, the expected outcome along the task-unrelated plane was never in conflict with that in the task-related plane. For example, if a stimulus is expected on the same (right) side of the display (repeat; task-related plane), then regardless of whether the stimulus appeared above or below fixation, there was a quadrant of the display that would satisfy both the task-related and unrelated expectations (bottom- or top-right). This is because the expectation was either for repeat stimuli (micro) or balanced between repeat and alternate stimuli (meso). By contrast, in the macro condition, where stimuli could be expected to alternate, this produced trials on which the task-related and unrelated expectations conflicted, such that there was no location that could satisfy both. Accuracy along the task-unrelated plane was reduced in these instances where there was conflict. Asterisks indicate significant differences (** $p < .01$, *** $p < .001$). **e)** The difference (expected – unexpected) in pupil size as a function of time from stimulus presentation, along the task-related (TR) plane. Shaded region indicates $\pm$ SEM and the horizontal bar indicates a cluster corrected period of significant difference.

precision for these stimuli and, consistent with recent psychophysical work (27 [↗](#)), suggests that unexpected (alternating) events are prioritized in terms of both encoding fidelity and processing speed. Given that unexpected stimuli were responded to more slowly than expected stimuli, despite benefitting from sensory processing priority, this may indicate that the latter is supported by anticipatory motoric activity (28 [↗](#)).

Effects on binary judgements associated with micro temporal context, such as those described above, have been referred to as first-order sequential dependencies (16 [↗](#)). By contrast, those associated with continuous report tasks are typically referred to as serial dependence (5 [↗](#), 7 [↗](#)). We examined behavioural responses for continuous effects and found the stereotypical pattern of response biases associated with serial dependence, i.e., attraction towards the preceding stimulus.

Our corresponding EEG analyses revealed better decoding accuracy and precision for stimuli preceded by those that were different and a bias away from the previous stimulus. These results are consistent with finding that alternating stimuli are recalled more precisely. Further, while the repulsive pattern of biases is inconsistent with the observed behavioural attractive biases, it is consistent with recent work on serial dependence indicating an initial period of repulsion, followed by an attractive bias during the response period (20 [↗](#)–22 [↗](#)). These findings indicate that serial dependence and first-order sequential dependencies can be explained by the same underlying principle. Behavioural and neural recordings show a predisposition for expected outcomes, which manifest as faster, biased, behavioural responses. By contrast, unexpected outcomes are encoded earlier and more precisely, leading to increased recall fidelity for these events.

Meso-scale temporal context

At the meso scale, consistent with previous work (9 [↗](#), 10 [↗](#)), we found that short sequences of stimuli were responded to faster and more accurately when they were expected. Extending these findings, and matching the influence of micro-scale temporal context, we also found that responses were more precise for unexpected sequences (in Experiment 2, and non-significantly in Experiment 1). Pupillometry analyses confirmed that participants were surprised by stimuli that were categorized as unexpected, and EEG analyses revealed a strikingly similar pattern of results as was found at the micro scale. In particular, unexpected stimuli were encoded earlier and more robustly. The similarity between the influence at the micro and meso scale is particularly remarkable given that the influence of micro temporal context was controlled for in the meso-scale analyses.

Theoretical and empirical work has implicated both motor and sensory priming in the modulation of response time and accuracy associated with higher-order sequential dependencies (28 [↗](#)–30 [↗](#)). Decoupling the motoric component of experimental designs used to probe the phenomenon has been an obstacle to assessing how sensory processing is influenced by meso-scale temporal context. In the EEG experiment that was re-analyzed here, participants viewed serially presented stimuli and were tasked with reporting the number of targets after 30 presentations. Thus, there were no motor actions generated during the recording period used to perform the analysis and our results can only be attributed to differences in visual processing between sequences. Indeed, the timing of the improved encoding of unexpected sequences, like that for micro temporal context, can only be accounted for by a mechanism that prioritizes the encoding speed and fidelity of stimuli during the initial feedforward cascade of sensory processing.

Macro temporal context

Results from our second experiment, in which biases in the probability of repeat and alternate stimuli were introduced, showed that responses are faster and more accurate for expected stimuli, and that recall precision is better for unexpected stimuli. Pupillometry analysis confirmed that participants were surprised by unexpected stimuli. We replicated these results in a third

experiment, demonstrating that the same outcome for stimuli of fixed duration. These findings parallel those observed at the micro and meso scales and point to two canonical adaptive mechanisms that operate across a wide range of temporal contexts to support a common sensory outcome.

The role of attention

At each scale, we found that attention (operationalized through task-relatedness) amplified the influence of temporal context for response time and task accuracy, but not precision. These findings suggest that perception and behaviour are modulated by expectation through two distinct mechanisms. One mechanism is sensitive to top-down goals (attention) and supports rapid responses to expected stimuli, while the other mechanism automatically detects mismatches between expected and actual outcomes and prioritizes the encoding speed and fidelity of unexpected stimuli. In line with this, some studies have reported interactions between attention and predictive mechanisms (31, 32), while others have shown that oddball event-related potentials can be detected in coma patients (33).

Consistent with previous work (17), we found that the influence of attention on the neural representation only emerged after ~200 ms. By contrast, we found that effects of expectation, for both micro- and meso-scale temporal context, were present during the earliest moments of cortical sensory processing, i.e., during the feedforward cascade. These findings suggest that predictive mechanisms are operational during feedforward processing, while attention modulates sensory processing at a later stage.

Predictive coding

Predictive coding theories argue that *prediction errors* are generated when bottom-up sensory inputs deviate from top-down expectations (3). There is broad consensus that prediction errors are associated with increased neural activation (13, 34–38), which is typically observed during the initial processing cascade in sensory cortices associated with the unexpected stimulus (39, 40), but has also been reported in subcortical regions (24). By contrast, whether this modulation of neural activation results from ‘dampening’ or ‘sharpening’ of the neurons tuned to expected features remains highly contested (3, 13–15, 41). Whether through increased selectively (sharpening) or reduced sensitivity (dampening), both accounts predict attenuated activation in response to expected events. However, according to the sharpening model, expected events are represented more precisely, while the dampening model posits the opposite, i.e., unexpected events are more precisely represented. At every scale of temporal context and across behavioural and neural recordings, we consistently found that unexpected events were encoded more precisely than expected events. These findings provide compelling evidence that predictive mechanisms reduce the sensitivity of neurons tuned to expected features, likely in order to reduce metabolic energy expenditure (3), rather than increasing their selectively.

The expectation of stimuli shaped their neural representation in a robust and striking manner. In particular, the peak in classification accuracy from 50–150 ms following stimulus onset, which represents the initial stage of (feedforward) cortical representation, was almost entirely accounted for by stimulus expectation such that expected stimuli were only minimally represented during this period. This modulation cannot be attributed to adaptation, as we found the same result at the meso scale, where the change in location between subsequent stimuli was matched between conditions. Further, we found an attractive bias during this period, as opposed to the repulsive bias associated with adaptation. There was relatively weak evidence for a difference between the neural representation of expected and unexpected stimuli following this initial peak. Given that unexpected stimuli were recalled more precisely in the behavioural task, this provides further support to the notion that recall of these stimuli primarily reflects neural activity during this initial stage of sensory processing; however, more work is needed to understand how this relationship changes with increasing stimulus complexity. These findings provide empirical

evidence that predictive mechanisms operate to reduce the feedforward sensory activity associated with expected stimuli in a manner that reduces their representational fidelity (i.e., dampening/cancellation).

Prioritizing the speed and fidelity with which unexpected events are encoded likely serves to rapidly and precisely update internal predictive models when there is a mismatch between predictions and incoming sensory information. By contrast, expected events may benefit from anticipatory motoric activation, which facilitates rapid behavioural responses, and do not require encoding prioritization.

Limitations and future directions

One potential limitation of the current study is the categorization of temporal scales according to events, independent of the influence of event duration. While this simplification of time supports comparison between different phenomena associated with each scale (e.g., serial dependence, sequential dependencies, statistical learning), future work could investigate the role of duration to provide a more comprehensive understanding of the mechanisms identified in the current study. Related to this, while the temporal scales applied here conveniently categorized known sensory phenomena, and partially correspond to iconic-, short-, and long-term memory, there are alternative ways of delineating time. For example, temporal scales could be defined simply as short- and long-term ([42](#)). However, this could obscure meaningful differences between phenomena associated with sensory persistence and short-term memory, or qualitative differences in the way that short-sequences of events are processed.

Another limitation of the current study is that the EEG recordings were collected in a separate experiment to the behavioural and pupillometry data. The stimuli and task were similar between experiments, but not identical. For example, the EEG experiment employed coloured arc stimuli presented at a constant rate of ~ 3.3 Hz and participants were tasked with counting the number of stimuli presented at a target location. By contrast, in the behavioural experiment, participants viewed white blobs presented at an average rate of ~ 2.8 Hz and performed a binary spatial task coupled with an infrequent reproduction task. An advantage of this was that the sensory responses to stimuli in the EEG recordings were not conflated with motor responses; however, future work combining these measures in the same experiment would serve as a validation for the current results.

Conclusion

Past sensory information shapes perception of the present. Multiple research areas have emerged to investigate the isolated influence of temporal context at different scales. In a single experimental paradigm, we compared the influence of past stimuli at micro, meso, and macro temporal scales, and found two common rules that characterized the relationship between the past and present across all ranges. While the neural mechanisms that implement the influence of temporal context across different scales may vary (e.g., tonic inhibition/excitation, synaptic weight), they appear to promote a two unifying outcomes: 1) rapid motor responses to expected events, and 2) prioritized encoding speed and fidelity of unexpected events.

Methods

Participants

Eighty neurotypical human adults participated in Experiments 1 and 2 (mean $\pm$ standard deviation age, Experiment 1: 20.9 $\pm$ 2.9 years, 14 males, 65 females, 1 non-binary, Experiment 2: 21.0 $\pm$ 5.8 years, 18 males, 62 females), and 40 neurotypical human adults participated in Experiment 3 (24.7 $\pm$ 8.3 years, 10 males, 30 females. The data of one participant was omitted from Experiment 1

because they were unable to complete the experiment. Sample size was informed by previous studies using similar psychophysical methods (27 [↗](#)). Participants were recruited from The University of Sydney and had normal or corrected-to-normal vision (assessed using a standard Snellen eye chart). All participants were naïve to the aims of the experiment and gave informed written consent. The experiment was approved by The University of Sydney Human Research Ethics Committee.

Apparatus

The experiment was conducted in a dark acoustically shielded room. The stimuli were presented on a 41.5-inch ASUS ROG Swift OLED monitor with 3840 x 2160 resolution and a refresh rate of 120 Hz. Viewing distance was maintained at 1 m using a chin and head rest. Stimuli were generated in MATLAB v2020a (The MathWorks, Inc., Matick, MA) using Psychophysics Toolbox (43 [↗](#), 44 [↗](#)) v3.0.18.13 (see <http://psychtoolbox.org/> [↗](#)). Gaze direction and pupil diameter were recorded monocularly (right eye) at 1 kHz using an EyeLink 1000 (SR Research Ltd., Ontario, Canada).

Stimuli, task, and procedure

The stimuli comprised white Gaussian blobs (standard deviation=0.3°, contrast=0.5) positioned 4° of visual angle from fixation (location randomly selected between 0-360°) on a black background. A centrally positioned white fixation dot (radius=0.25°) was presented to reduce eye movements. Stimuli were presented until the participant responded. Half the participants were instructed to use the “z” and “/” keys to indicate whether the stimulus appeared on the left or right of fixation, and the other half used the “b” and “t” keys to indicate whether it appeared above or below fixation; the keys were selected to map the relative positions on the keyboard to the orientation of the task. Participants performed 15 to 20 blocks of 100 trials (~90 min), receiving feedback on their response time and task accuracy at the end of each block. On 10% of randomly selected trials, after participants responded, the target blob was replaced by a green blob at a random location (same size as target blob), and participants were instructed to use the response keys to rotate the location of the green blob around to the location of the previous target blob and press the “spacebar” key to confirm their response.

Experiment 2 was the same as Experiment 1, except that biases were introduced to the location of the target blob. In particular, sequentially presented target stimuli were more likely to be presented on the same side (75%; repeat) than on the other side (25%; alternate), or vice versa, along each of the task reference planes (left/right and above/below). Thus, a 2 (task reference plane) × 2 (left/right bias) × 2 (above/below bias) design was used and the eight conditions were counterbalanced across participants. On each trial, based on the location of the target on the previous trial and the unique bias condition, the quadrant of the next target was probabilistically determined and then the location was randomly selected within the quadrant. Critically, the biases manipulated the probability of repeat and alternate stimuli, not the spatial location (e.g., left or right). Experiment 3 was the same as Experiment 2, but stimuli were presented for a fixed duration (0.2 s). This duration was selected because it was below the normal range of response times and was intended to allow the stimulus to be presented for the full duration before participants responded.

Eye tracking

Pupil size recordings were epoched to between -0.5 to 1.5 s around target stimulus presentation. Pupillometry data were preprocessed by removing blinks (0.1 s buffer window), removing outliers (median absolute deviation>2.5), interpolated, bandpass filtered with a first-order Butterworth filter (highpass=15.6 Hz, lowpass=500 Hz), z-scored, baselined to the 0.5 s period before stimulus presentation, and downsampled to 125 Hz, respectively. Due to hardware issues, pupil data were not collected from one participant in Experiment 1, two in Experiment 2, and three in Experiment 3.

Behavioural analyses

Trials were grouped according to whether stimuli repeated (the previous stimulus was presented on the same side) or alternated (the previous stimulus was presented on the opposite side) according to three reference planes: task-related, task-unrelated, and stimulus-related. The task-related plane mapped onto the participants' task condition (left-right or above-below) and the task-unrelated plane was orthogonal to the task-related plane. For example, if a participant was assigned the left-right task condition, the task-related plane was defined by the vertical meridian and the task-unrelated plane was defined by the horizontal meridian. The stimulus-related plane was defined by the line bisecting the fixation point at an angle orthogonal to the stimulus location on the previous trial, and thus changed over time.

For micro analyses, the difference in responses to repeat and alternate stimuli was compared. For meso analyses, trials were further sorted into eight pairs of unique sequences, based on the (repeat/alternate) identity of the previous five trials ([Fig. 4a](#)). For macro analyses in Experiment 2 and 3, repeat and alternate trials were grouped according to whether they were expected (75% probability) or unexpected (25% probability), based on participants' experimental condition. For all analyses, sequences that were disrupted by a reproduction task or block advancement were omitted. For each participant, after removal of trials where response time exceeded 2 s, median response times and average task accuracy were calculated.

From the reproduction task, we sought to estimate participant's recall precision. It is likely that on some trials participants failed to encode the target and were forced to make a response guess. To isolate the recall precision from guess responses, we used mixture modelling to estimate the precision and guess rate of reproduction responses, based on the concentration (κ) and height of von Mises and uniform distributions, respectively ([45](#)). The κ parameter of the von Mises distribution reflects its width, which indicates the clustering of responses around a common location. For the serial dependency analysis, we calculated bias and precision by binning reproduction responses according to the angular offset between the current and previous target and then performing mixture modelling to estimate the mean (bias) and κ (precision) parameters of the von Mises distribution. Precision values were positively skewed across participants, so a logarithmic transformation was applied to normalize their distribution ($\log_{10} \kappa$). For serial dependence analyses, the location (μ) of the von Mises distribution was used to estimate response bias.

EEG

EEG data from a previously published dataset were re-analyzed ([17](#)). The stimuli comprised coloured arcs (inner ring, 0.25°; outer ring, 4.25°) extending 45° polar angle on a mid-grey background. Trials consisted of 30 arc stimuli (location and colour randomly selected between 0–360°) presented for 0.2 s each, separated by a blank 0.1 s inter-stimulus-interval. Participants were instructed to indicate the number of target stimuli that were presented during the preceding trial. At the beginning of each block, target stimuli were identified as either those appearing at a defined location or colour, counterbalanced across the session. EEG data from blocks in which the target was defined by spatial location were included in micro and meso linear discriminant analyses; all data were included in the sequential dependency forward encoding analysis.

The EEG recordings were digitised at 1024 Hz sampling rate with a 24-bit A/D conversion. The 64 active scalp Ag/AgCl electrodes were arranged according to the international standard 10–20 system for electrode placement([46](#)) using a nylon head cap and with an online reference of FCz. Offline EEG pre-processing was performed using EEGLAB v2021.1([47](#)). The data were initially subjected to a 0.1 Hz highpass filter to remove slow baseline drifts and a 45 Hz lowpass filter to remove high-frequency noise/artifacts before being downsampled to 512 Hz. Data were then re-referenced to the common average before being epoched into segments around each stimulus

(-0.125 s to 0.5 s from the stimulus onset). We only included the parietal, parietal-occipital, and occipital sensors in the analyses to *i*) reduce the influence of signals produced by eye movements, blinks, and non-sensory cortices, *ii*) for consistency with similar previous decoding studies (17 [48](#), 49 [48](#)), and *iii*) to improve decoding accuracy by restricting sensors to those that carried spatial position information (**Supplementary Fig. 1a** [48](#)).

Neural Decoding

For micro and meso analyses, EEG recordings (epochs) were grouped according to which side of the task-related or unrelated reference plane they were presented. As the task in the EEG experiment was to detect stimuli presented at a specific location, rather than perform a binary judgement (e.g., left or right of fixation), we constructed task boundaries based on the spatial coordinates that were indicative to the target detection task. Thus, the task-related reference plane was defined by the line bisecting the fixation point at an angle orthogonal to the target location, because this plane was most informative of whether a stimulus was a target. By contrast, the task-unrelated plane was the line orthogonal to the task-related line, as this plane was uninformative of whether a stimulus was a target. Ten-fold cross-validation linear discriminant analysis was performed with the MATLAB *classify* function (discriminant function, diaglinear) to calculate the two-way classification accuracy separately at each time point and reference plane. Classification accuracy was then separately assessed for repeat and alternate trials (micro analysis) and the eight pairs of five-event sequences (meso analysis).

As shown in **Supplementary Figure 2** [48](#), we found that there were differences in classification accuracy for repeat and alternate stimuli in the EEG data, even when stimulus labels were shuffled. This is likely due to temporal autocorrelations within the EEG data that are unrelated to the decoded stimulus dimension. This signal causes the decoder to classify temporally proximal stimuli as the same class, leading to a bias towards repeat classification. For example, in general, the EEG signal during trial one will be more similar to that during trial two than during trial ten, because of low frequency trends in the recordings. If the decoder has been trained to classify the signal associated with trial one as a leftward stimulus, then it will be more likely to classify trial two as a leftward stimulus too. These autocorrelations are unrelated to stimulus features; thus, to isolate the influence of stimulus-specific temporal context, we subtracted the accuracy produced by shuffling the stimulus labels from the unshuffled accuracy. As a further test of this explanation, we performed the same classification analysis on the data after applying a high-pass filter of 0.7 Hz.

For serial dependency analyses, we used an inverted modelling approach to reconstruct either the location or Δ location (angular distance from previous stimulus) of the stimuli (50 [48](#)); note, the results of Δ location analyses are shown in **Supplementary Figure 1** [48](#). A theoretical (forward) model was nominated that described the measured activity in the EEG sensors given the location/ Δ location of the presented stimulus. The forward model was then used to obtain the inverse model that described the transformation from EEG sensor activity to stimulus location/ Δ location. The forward and inverse models were obtained using a ten-fold cross-validation approach.

Similar to previous work (26 [48](#), 51 [48](#), 52 [48](#)), the forward model comprised five hypothetical channels, with evenly distributed idealized location/ Δ location preferences between 0° and 360°. Each channel consisted of a half-wave rectified sinusoid raised to the fifth power. The channels were arranged such that a tuning curve of any location/ Δ location preference could be expressed as a weighted sum of the five channels. The observed EEG activity for each presentation could be described by the following linear model:

$$\mathbf{B} = \mathbf{WC} + \mathbf{E} \quad (1)$$

where $\mathbf{B}$ indicates the (m sensors $\times$ n presentations) EEG data, $\mathbf{W}$ is a weight matrix (m sensors $\times$ 5 channels) that describes the transformation from EEG activity to stimulus location/ Δ location, $\mathbf{C}$ denotes the hypothesized channel activities (5 channels $\times$ n presentations), and $\mathbf{E}$ indicates the residual errors.

To compute the inverse model, we estimated the weights that, when applied to the data, would reconstruct the underlying channel activities with the least error. In line with previous magnetencephalography work (53, 54), when computing the inverse model, we deviated from the forward model proposed by Brouwer and Heeger (2009) (51) by taking the noise covariance into account to optimize it for EEG data, given the high correlations between neighbouring sensors. We then estimated the weights that, when applied to the data, would reconstruct the underlying channel activities with the least error. Specifically, $\mathbf{B}$ and $\mathbf{C}$ were demeaned such that their average over presentations equalled zero for each sensor and channel, respectively. The inverse model was then estimated using a subset of data selected through cross-fold validation. The hypothetical responses of each of the five channels were calculated from the training data, resulting in the response row vector $\mathbf{c}_{train,i}$ of length n_{train} presentations for each channel i . The weights on the sensors $\mathbf{w}_i$ were then obtained through least squares estimation for each channel:

$$\mathbf{w}_i = \mathbf{B}_{train} \mathbf{c}_{train,i}^T (\mathbf{c}_{train,i} \mathbf{c}_{train,i}^T)^{-1} \quad (2)$$

where n_{train} indicates the (m sensors $\times$ n_{train} presentations) training EEG data. Subsequently, the optimal spatial filter $\mathbf{v}_i$ to recover the activity of the i th channel was obtained as follows (54):

$$\mathbf{v}_i = \frac{\tilde{\Sigma}_i^{-1} \mathbf{w}_i}{\mathbf{w}_i^T \tilde{\Sigma}_i^{-1} \mathbf{w}_i} \quad (3)$$

where $\tilde{\Sigma}_i$ is the regularized covariance matrix for channel i . Incorporating the noise covariance in the filter estimation leads to the suppression of noise that arises from correlations between sensors. The noise covariance was estimated as follows:

$$\hat{\Sigma}_i = \frac{1}{n_{train} - 1} \boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^T \quad (4)$$

$$\boldsymbol{\epsilon}_i = \mathbf{B}_{train} - \mathbf{w}_i \mathbf{c}_{train,i} \quad (5)$$

where n_{train} is the number of training presentations. For optimal noise suppression, we improved this estimation by means of regularization by shrinkage using the analytically determined optimal shrinkage parameter (54), yielding the regularized covariance matrix $\tilde{\Sigma}_i$.

For each presentation, we decoded location and Δ location by converting the channel responses to polar form:

$$z = \mathbf{c} \cdot e^{2i\phi} \quad (6)$$

and calculating the estimated angle:

$$\hat{\theta} = \frac{\arg(z)}{2} \quad (7)$$

where $\mathbf{c}$ is a vector of channel responses and ϕ is the vector of angles at which the channels peak. Decoding accuracy, which represents the similarity of the decoded location/colour to the presented location and Δ location (53), was expressed by projecting the mean resultant (presentations

averaged across 12 evenly distributed Δ location bins) of the difference between decoded and arc location onto a vector with 0° :

$$\begin{aligned}\hat{r}_\theta &= \text{Re}[\bar{R}], \quad \bar{R} \\ &= \frac{1}{n} \sum_{j=1}^n \exp(i(\hat{\theta}_j - \theta))\end{aligned}\quad (8)$$

Precision was estimated by calculating the angular deviation (55) of the decoded orientations within each orientation bin:

$$\hat{\sigma}_\theta = \sqrt{2(1 - |\bar{R}|)} \quad (9)$$

and normalized, such that values ranged from 0 to 1, where 0 indicates a uniform distribution of decoded orientations across all orientations (i.e., chance-level decoding) and 1 represents perfect consensus among decoded orientations:

$$\hat{p}_\theta = 1 - \frac{2\hat{\sigma}_\theta}{\sqrt{2}} \quad (10)$$

Bias was estimated by computing the circular mean of angular difference between the decoded and presented orientation:

$$\hat{b}_\theta = \arg(\bar{R}) \quad (11)$$

As shown in **Supplementary Figure 3**, we found the same general temporal dependencies in the decoding accuracy computed using inverted encoding that were found using linear discriminant classification. However, as a baseline correction would not have been appropriate or effective for the parameters decoded with this approach, we instead used a high-pass filter of 0.7 Hz to remove the confound, while being cautious about interpreting the timing of effects produced by this analysis due to the temporal smear introduced by the filter.

Statistical analyses

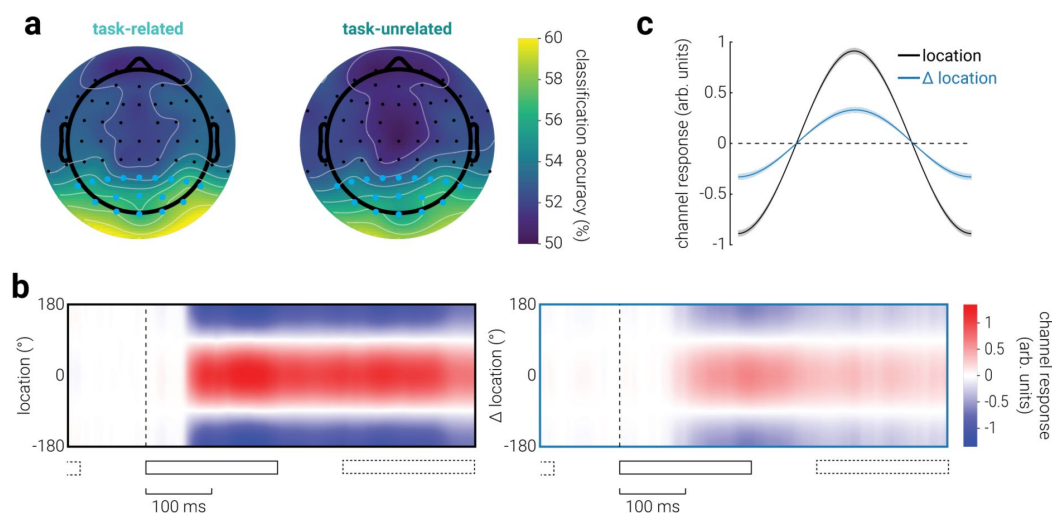
Statistical analyses were performed in MATLAB v2020a and CircStat Toolbox v1.12.0.0 (56). Prior to statistical testing of behavioural estimates, within-participant average values of response time, task accuracy, and precision estimates >1.5 times the interquartile range of the group distribution were removed. Non-parametric inferential tests (Wilcoxon signed rank test (57) and Friedman's ANOVA(58)) were used to assess the significance of paired differences and main effects. For Wilcoxon signed rank tests, the rank-biserial correlation (r) was calculated as an estimate of effect size, where 0.1, 0.3, and 0.5 indicate small, medium, and large effects, respectively (59). For Friedman's ANOVA tests, Kendal's W was calculated as an estimate of effect size, where 0.1, 0.3, and 0.5 indicate small, medium, and large effects, respectively (60). Levene's test of homogeneity was used to compare the difference in variance between conditions (61). For pupillometry and linear discriminant analyses, one-dimensional cluster correction was applied to remove spurious significant differences (62). For inverted modelling analyses of location and Δ location as a function of Δ location and time, a two-dimensional circular-linear cluster correction was applied. First, at each Δ location and/or time point, the effect size of the dependent variable (e.g., decoding accuracy) was calculated. Next, we calculated the summed value of these statistics (separately for positive and negative values) within contiguous featural and/or temporal clusters of significant values. We then simulated the null distribution of the maximum summed cluster values using permutation ($n=5000$) of the sign or condition labels, for single and paired t -test comparisons, respectively, from which we derived the 95% percentile threshold value. Clusters identified in the data with a summed effect-size value less than the

threshold were considered spurious and removed. Prior to cluster analysis, we smoothed the decoding accuracy estimates in the inverted modelling analyses along the feature dimension using a uniform kernel (size, 4).

Data and Code Availability

The data and analysis code generated in this study have been deposited in the following OSF database: <https://osf.io/9qfdk/>

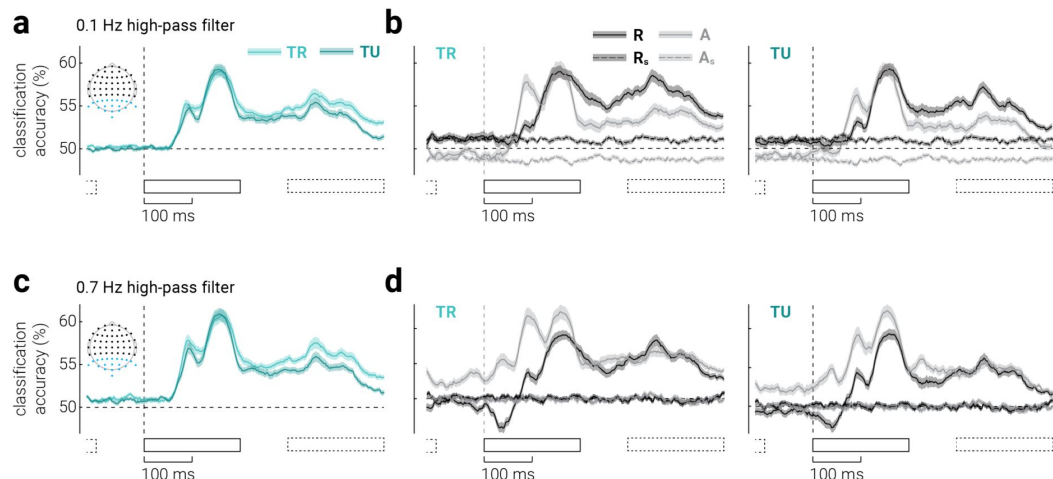
Supplemental Information



Supplementary Figure 1.

Decoding stimulus location and Δ location from EEG recordings.

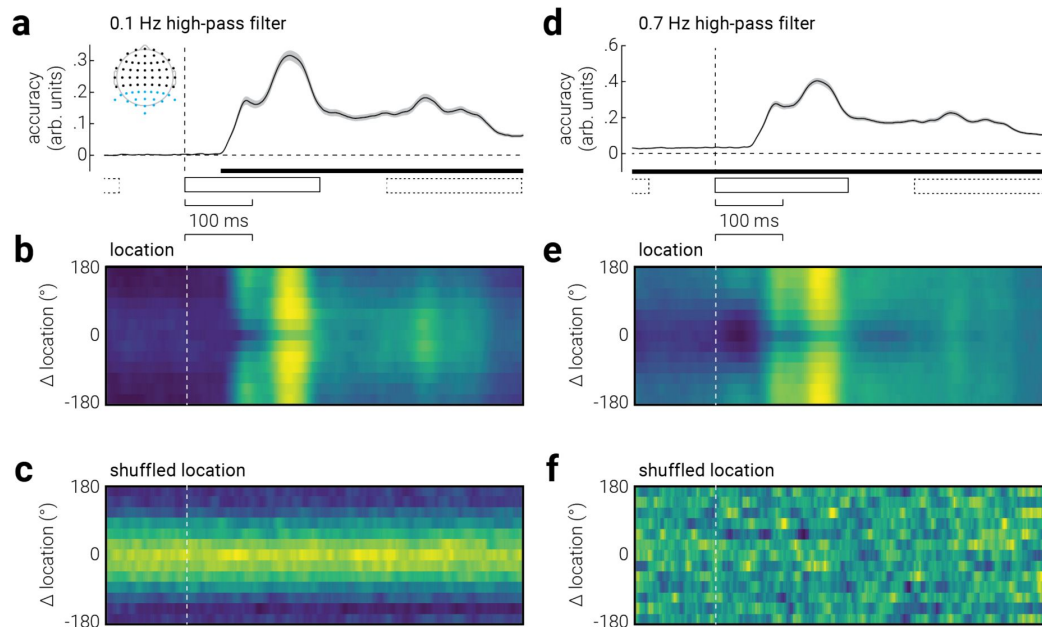
a) Topographic representation of task-related and unrelated stimulus location information produced using linear discriminant analysis to classify stimulus location, separately for each sensor, with time as the multivariate dimension. Blue dots indicate the sensors that were subsequently used in all other EEG analyses. **b)** Channel responses for location and Δ location, produced by computing the dot product between the inverted model channel responses and the forward model, as a function of time. **c)** The average channel response for location and Δ location between 0-500 ms following stimulus onset.



Supplementary Figure 2.

Removal of general micro temporal dependencies in EEG responses.

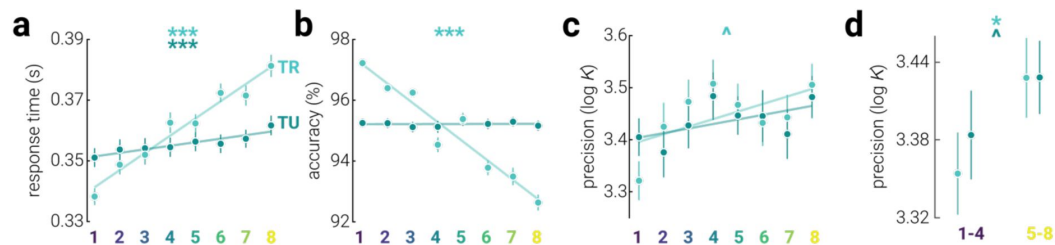
We found that there were differences in classification accuracy for repeat and alternate stimuli in the EEG data, even when stimulus labels were shuffled. This is likely due to temporal autocorrelation within the EEG data due to low frequency signal changes that are unrelated to the decoded stimulus dimension. This signal trains the decoder to classify temporally proximal stimuli as the same class, leading to a bias towards repeat classification. For example, in general, the EEG signal during trial one is likely to be more similar to that during trial two than during trial ten, because of low frequency trends in the recordings. If the decoder has been trained to classify the signal associated with trial one as a leftward stimulus, then it will be more likely to classify trial two as a leftward stimulus too. These autocorrelations are unrelated to stimulus features; thus, to isolate the influence of stimulus-specific temporal context, we subtracted the classification accuracy produced by shuffling the stimulus labels from the unshuffled accuracy (as presented in [Figure 2e, f](#)). We confirmed that using a stricter high-pass filter (0.7 Hz) removes this artifact, as indicated by the equal decoding accuracy between the two shuffled conditions. However, the stricter high-pass filter temporally smears the stimulus-related signal, which introduces other (stimulus-related) artifacts, e.g., above-chance decoding accuracy prior to stimulus presentation, that are larger and more complex, i.e., changing over time. Thus, we opted to use the original high pass filter (0.1 Hz) and apply a baseline correction. **a)** The uncorrected classification accuracy along task related and unrelated planes. Note that these results are the same as the corrected version shown in [Figure 2e](#), because the confound is only apparent when accuracy is grouped according to temporal context. **b)** Same as **(a)**, but split into repeat and alternate stimuli, along (left) task-related and (right) unrelated planes. Classification accuracy when labels are shuffled is also shown. Inset in **(a)** shows the EEG sensors included in the analysis (blue dots). **(c, d)** Same as **(a, b)**, but on data filtered using a 0.7 Hz high-pass filter. Black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). Shaded regions indicate $\pm$ SEM.



Supplementary Figure 3.

Removal of general temporal dependencies in EEG responses for inverted encoding analyses.

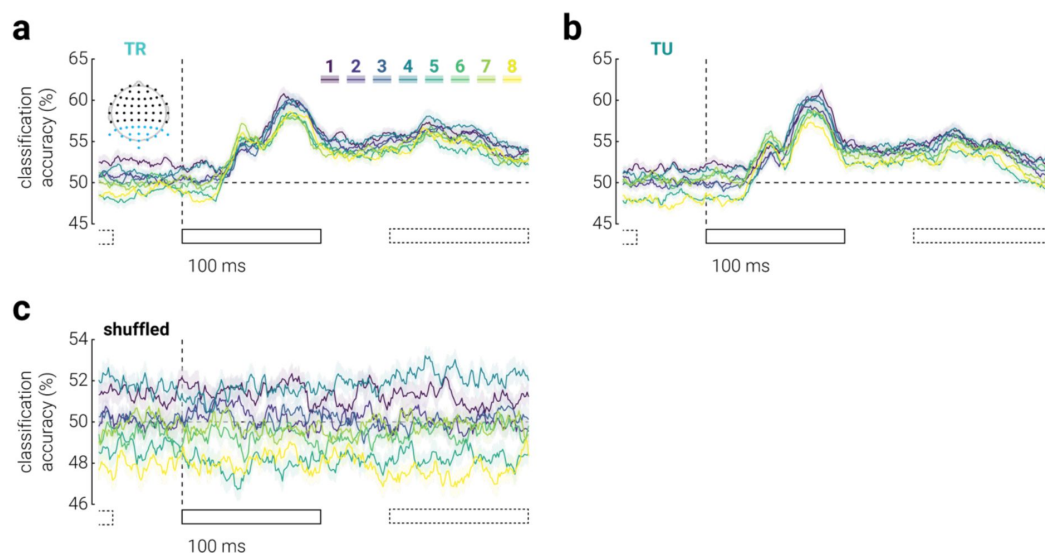
As described in *Methods - Neural Decoding*, we used inverted encoding modelling of EEG recordings to estimate the decoding accuracy, precision, and bias of stimulus location. Just as in the linear discriminant classification analysis, we also found the influence of general temporal dependencies in the results produced by the inverted encoding analysis. In particular, there was increased decoding accuracy for targets with low Δ location. This was weakly evident in the period prior to stimulus presentation, but clearly visible when the labels were shuffled. These results are mirror those from the classification analysis, albeit in a more continuous space. However, whereas in the classification analysis it was straightforward to perform a baseline correction to remove the influence of general temporal dependency, the more complex nature of the accuracy, precision, and bias parameters over the range of time and Δ location makes this approach less appropriate. For example, the bias in the shuffled condition ranged from -180° to 180° , which when subtracted from the bias in the unshuffled condition would produce an equally spurious outcome, i.e., the equal opposite of this extreme bias. Instead for the inverted encoding analysis, we used the data high-pass filtered at 0.7 Hz. As with the classification analysis, this significantly reduced the influence of general temporal dependencies, as indicated by the results of the shuffled data analysis, but it also temporally smeared the stimulus-related signal, resulting in above chance decoding accuracy prior to stimulus onset. However, we were primarily interested in the pattern of accuracy, precision, and bias as a function of Δ location, and less concerned with the precise temporal dynamics of these changes. Thus, this was the more suitable approach to removing the general temporal dependencies in the inverted encoding analysis and the one that is presented in **Figure 3**. (a) Decoding accuracy as a function of time for the EEG data filtered using a 0.1 Hz high-pass filter. Inset shows the EEG sensors included in the analysis (blue dots), and black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). (b, c) The same as (a), but as a function of time and Δ location for (b) the original data and (c) data with shuffled labels. (d-f) Same as (a-c), but for data filtered using a 0.7 Hz high-pass filter. Shaded regions in (a, d) indicate $\pm$ SEM. Horizontal bars in (a, d) indicate cluster corrected periods of significance; note, all time points in (d) were significantly above chance. Note, the temporal abscissa is vertically aligned across plots (a-c & d-f).



Supplementary Figure 4.

Meso-scale temporal context effects in Experiment 2.

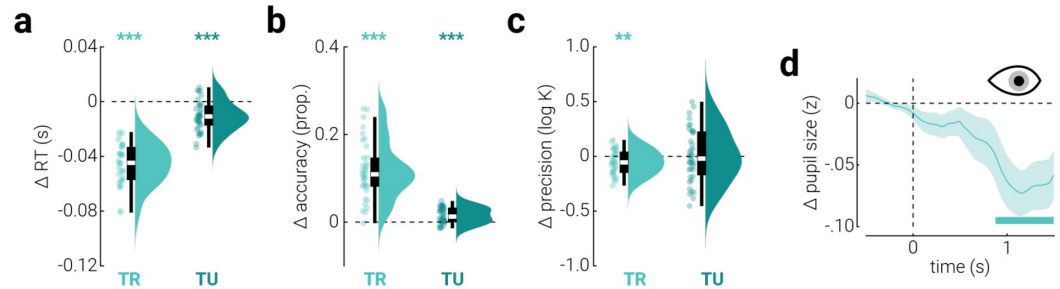
a) Response time, **(b)** task accuracy, and **(c)** precision as a function of sequence, across task-related and unrelated reference planes. Note, lower numbers on the abscissa are associated with more expected sequences. **d)** Same as **(c)**, but pooled across the four most expected (1-4) and unexpected sequences (5-8). Asterisks indicate significant correlations with sequence order (* $p < .05$, ** $p < .01$, *** $p < .001$), hats indicate marginal significance ($^{\wedge}p < .06$). Error bars indicate $\pm$ SEM.



Supplementary Figure 5.

Removal of general meso temporal dependencies in EEG responses.

To remove any potential general sequential dependencies from the meso-scale classification analysis of EEG recordings, we performed the same subtraction method as used in the micro-scale analysis. **(a)** Classification accuracy of stimuli presented on different sides of task-related planes as a function of time, for each of the eight sequences, from re-analysis of previously published EEG data (17). **(b)** Same as **(a)**, but for the task-unrelated plane. **(c)** Same as **(a)**, but with shuffled labels. **Figure 4f, g** shows **(a, b)** after removing shuffled accuracy **(c)**. Inset in **(a)** shows the EEG sensors included in the analysis (blue dots). Black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). Shaded regions indicate $\pm$ SEM.



Supplementary Figure 6.

Macro-scale temporal context effects in Experiment 3.

a-c) The difference (expected – unexpected) in **(a)** response time, **(b)** task accuracy, and **(c)** reproduction precision between expected and unexpected stimuli, along task-related (TR) and unrelated (TU) planes. Asterisks indicate significant differences (** $p < .01$, *** $p < .001$). **d)** The difference (expected – unexpected) in pupil size as a function of time from stimulus presentation, along the task-related (TR) plane. Shaded region indicates $\pm$ SEM and the horizontal bar indicates a cluster corrected period of significant difference.

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Author information

Kacie Lee

School of Psychology, The University of Sydney, Camperdown, Australia

Reuben Rideaux

School of Psychology, The University of Sydney, Camperdown, Australia, Queensland Brain Institute, The University of Queensland, St Lucia, Australia

ORCID iD: [0000-0001-8416-005X](https://orcid.org/0000-0001-8416-005X)

For correspondence: reuben.rideaux@sydney.edu.au

Editors

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Tessa Dekker

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Reviewer #1 (Public review):

Summary:

This paper addresses an important and topical issue: how temporal context - at various time scales - affects various psychophysical measures, including reaction times, accuracy and localization. It offers interesting insights, with separate mechanisms for different phenomena, which are well discussed.

Strengths:

The paradigm used is original and effective. The analyses are rigorous.

Comments on revised version:

I think the authors have dealt adequately with my issues, none of which were fundamental.

<https://doi.org/10.7554/eLife.106614.2.sa2>

Reviewer #2 (Public review):

Summary:

This study investigates the influence of prior stimuli over multiple time scales in a position discrimination task, using pupillometry data and a reanalysis of EEG data from an existing dataset. The authors report consistent history-dependent effects across task-related, task-unrelated, and stimulus-related dimensions, observed across different time scales. These effects are interpreted as reflecting a unified mechanism operating at multiple temporal levels, framed within predictive coding theory.

Strengths:

The authors have done a good job in their revision, clarifying important points and stating the limitations of the study clearly.

I also think they made a valid effort to address and correct issues arising from the temporal dependency confound, although I still wonder whether the best approach would have been to design an experiment in a way that avoided this confound in the first place.

Overall, this is a substantially improved version, and I particularly appreciate the clarification and correction regarding the direction of the bias in the EEG data (repulsive rather than attractive).

Weaknesses:

These are now relatively minor points.

I believe this latter aspect, the repulsive bias, may deserve further discussion, especially in relation to their behavioral findings and, in particular, to earlier work proposing multi-stage frameworks of serial dependence, where low-level repulsion interacts with attractive biases at higher-level stages (Fritsche et al., 2020; Pascucci et al., 2019; Sheehan & Serences, 2022). The authors may also consider to cite some key reviews on serial dependence that discuss both repulsion and attraction in forced-choice and reproduction tasks (Manassi et al., 2023; Pascucci et al., 2023).

Related to this, after finding the opposite pattern, is the sentence in line 472-473 ("Further, we found an attractive...") and the related argument still valid?

Regarding my earlier point about former line 197 and Figure 3b,c: what I noticed-similar to the patterns reported in the studies I referenced-is that the data cannot be simply described as showing faster and more accurate responses for small deltas. Responses also appear faster and more accurate for very large deltas, with performance being worse in between. Indeed, as the authors state: "The peak in precision for large Deltas locations is consistent with alternate events being encoded more precisely, while the peak for small offsets may be explained by the attractive bias towards the previous target." I wonder whether it is necessary, or unequivocally supported by the data, to hypothesize two separate mechanisms here. An alternative could be interference effects between consecutive stimuli that are neither identical nor completely different-making the previous one more likely to interfere with the current stimulus representation.

Finally, this is definitely a minor point, but I still find the reply to my comment about the prediction of stable retinal input rather speculative. Such a prediction would seem more plausible in world-centered coordinates.

<https://doi.org/10.7554/eLife.106614.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) The manuscript is quite dense, with some concepts that may prove difficult for the non-specialist. I recommend spending a few more words (and maybe some pictures) describing the difference between task-relevant and task-irrelevant planes. Nice technique, but not instantly obvious. Then we are hit with "stimulus-related", which definitely needs some words (also because it is orthogonal to neither of the above).

We agree that the original description of the planes was too terse and have expanded on this in the revised manuscript.

Line 85 - To test the influence of attention, trials were sorted according to two spatial reference planes, based on the location of the stimulus: task-related and task-unrelated (Fig. 1b). The task-related plane corresponded to participants' binary judgement (Fig 1b, light cyan vertical dashed line) and the task-unrelated plane was orthogonal to this (Fig 1b, dark cyan horizontal dashed line). For example, if a participant was tasked with performing a left-or-right of fixation judgement, then their task-related plane was the vertical boundary between the left and right side of fixation, while their task-unrelated plane was the horizontal boundary. The former (left-right) axis is relevant to their task while the latter (top-bottom) axis is orthogonal and task irrelevant. This orthogonality can be leveraged to analyze the same data twice (once according to the task-related plane and again according to the taskunrelated plane) in order to compare performance when the relative location of an event is either task relevant or irrelevant.

Line 183 - whereas task planes were constant, the stimulus-related plane was defined by the location of the stimulus on the previous trial, and thus varied from trial to trial. That is, on each trial, the target is considered a repeat if it changes location by $< |90^\circ|$ relative to its location on the previous trial, and an alternate if it moves by $> |90^\circ|$.

(2) While I understand that the authors want the three classical separations, I actually found it misleading. Firstly, for a perceptual scientist to call intervals in the order of seconds (rather than milliseconds), "micro" is technically coming from the raw prawn. Secondly, the divisions are not actually time, but events: micro means one-back paradigm, one event previously, rather than defined by duration. Thirdly, meso isn't really a category, just a few micros stacked up (and there's not much data on this). And macro is basically patterns, or statistical regularities, rather than being a fixed time. I think it would be better either to talk about short-term and long-term, which do not have the connotations I mentioned. Or simply talk about "serial dependence" and "statistical regularities". Or both.

We agree that the temporal scales defined in the current study are not the only way one could categorize perceptual time. We also agree that by using events to define scales, we ignore the influence of duration. In terms of the categories, we selected these for two reasons: 1) they conveniently group previous phenomena, and 2) they loosely correspond to iconic-, short- and long-term memory. We agree that one could also potentially split it up into two categories (e.g., short- and long-term), but in general, we think any form of discretization will have limitations. For example, Reviewer 1 suggests that the meso category is simply a few micros stacked together. However, there is a rich literature on phenomena associated with sequences of an intermediate length that do not appear to be entirely explained by stacking micro effects (e.g., sequence learning and sequential dependency). We also find that when controlling for micro level effects, there are clear meso level effects. Also, by the logic that meso level effects are just stacked micro effects, one could also argue the same for macro effects. We don't think this argument is incorrect, rather we think it exemplifies the challenge of discretising temporal scales. Ultimately, the current study was aimed to test whether seemingly disparate phenomena identified in previous work could be captured by unifying principles. To this end we found that these categories were the most useful. However, we have included a "Limitations and future directions" section in the Discussion of the revised manuscript that acknowledges both the alternative scheme proposed by Reviewer 1, and the value of extending this work to consider the influence of duration (as well as events).

Line 488 - Limitations and future directions. One potential limitation of the current study is the categorization of temporal scales according to events, independent of the influence of

event duration. While this simplification of time supports comparison between different phenomena associated with each scale (e.g., serial dependence, sequential dependencies, statistical learning), future work could investigate the role of duration to provide a more comprehensive understanding of the mechanisms identified in the current study.

Related to this, while the temporal scales applied here conveniently categorized known sensory phenomena, and partially correspond to iconic-, short-, and long-term memory, they are but one of multiple ways to delineate time. For example, temporal scales could alternatively be defined simply as short- and long-term (e.g., by combining micro and meso scale phenomena). However, this could obscure meaningful differences between phenomena associated with sensory persistence and short-term memory, or qualitative differences in the way that shortsequences of events are processed.

(3) More serious is the issue of precision. Again, this is partially a language problem. When people use the engineering terms "precision" and "accuracy" together, they usually use the same units, such as degrees. Accuracy refers to the distance from the real position (so average accuracy gives bias), and precision is the clustering around the average bias, usually measured as standard deviation. Yet here accuracy is percent correct: also a convention in psychology, but not when contrasting accuracy with precision, in the engineering sense. I suggest you change "accuracy" to "percent correct". On the other hand, I have no idea how precision was defined. All I could find was: "mixture modelling was used to estimate the precision and guess rate of reproduction responses, based on the concentration (k) and height of von Mises and uniform distributions, respectively". I do not know what that means.

In the case of a binary decision, it seems reasonable to use the term "accuracy" to refer to the correspondence between the target state and the response on a task. However, we agree that while our (main) task is binary, the target is not and nor is the secondary task. We thank the reviewer for bringing this to our attention, as we agree that this will be a likely cause of confusion. To avoid confusion we have specifically referred to "task accuracy" throughout the revised manuscript.

With regards to precision, our measure of precision is consistent with what Reviewer 1 describes as such, i.e., the clustering of responses. In particular, the von Mises distribution is essentially a Gaussian distribution in circular space, and the kappa parameter defines the width of the distribution, regardless of the mean, with larger values of kappa indicating narrower (more precise) distributions. We could have used standard deviation to assess precision; however, this would incorrectly combine responses on which participants failed to encode the target (e.g., because of a blink) and were simply guessing. To account for these trials, we applied mixture modelling of guess and genuine responses to isolate the precision of genuine responses, as is standard in the visual working memory literature. However, we agree that this was not sufficiently described in the original manuscript and have elaborated on this method in the revised version.

Line 598 - From the reproduction task, we sought to estimate participant's recall precision. It is likely that on some trials participants failed to encode the target and were forced to make a response guess. To isolate the recall precision from guess responses, we used mixture modelling to estimate the precision and guess rate of reproduction responses, based on the concentration (k) and height of von Mises and uniform distributions, respectively (Bays et al., 2009). The k parameter of the von Mises distribution reflects its width, which indicates the clustering of responses around a common location.

(4) Previous studies show serial dependence can increase bias but decrease scatter (inverse precision) around the biased estimate. The current study claims to be at odds with that. But are the two measures of precision relatable? Was the real (random)

position of the target subtracted from each response, leaving residuals from which the inverse precision was calculated? (If so, the authors should say so..) But if serial dependence biases responses in essentially random directions (depending on the previous position), it will increase the average scatter, decreasing the apparent precision.

Previous studies have shown that when serial dependence is attractive there is a corresponding increase in precision around small offsets from the previous item (citations). Indeed, attractive biases will lead to reduced scattering (increased precision) around a central attractor. Consistent with previous studies, and this rational, we also found an attractive bias coupled with increased precision. To clarify, for the serial dependency analysis, we calculated bias and precision by binning reproduction responses according to the offset between the current and previous target and then performing the same mixture modelling described above to estimate the mean (bias) and kappa (precision) parameters of the von Mises distribution fit to the angular errors. This was not explained in the original manuscript, so we thank Reviewer 1 for bringing this to our attention and have clarified the analysis in the revised version.

Line 604 - For the serial dependency analysis, we calculated bias and precision by binning reproduction responses according to the angular offset between the current and previous target and then performing mixture modelling to estimate the mean (bias) and k (precision) parameters of the von Mises distribution.

(5) I suspect they are not actually measuring precision, but location accuracy. So the authors could use "percent correct" and "localization accuracy". Or be very clear what they are actually doing.

As explained in our response to Reviewer 1's previous comment, we are indeed measuring precision.

Reviewer #2 (Public review):

(1) The abstract should more explicitly mention that conclusions about feedforward mechanisms were derived from a reanalysis of an existing EEG dataset. As it is, it seems to present behavioral data only.

It is not clear what relevance the fact that the data has been analyzed previously has to the results of the current study. However, we do think that it is important to be clear that the EEG recordings were collected separately from the behavioural and eyetracking data, so we have clarified this in the revised abstract.

Line 7 - By integrating behavioural and pupillometry recordings with electroencephalographical recordings from a previous study, we identify two distinct mechanisms that operate across all scales.

(2) The EEG task seems quite different from the others, with location and color changes, if I understand correctly, on streaks of consecutive stimuli shown every 100 ms, with the task involving counting the number of target events. There might be different mechanisms and functions involved, compared to the behavioral experiments reported.

As stated above, we agree that it is important that readers are aware that the EEG recordings were collected separately to the behavioural and eyetracking data. We were forthright about this in the original manuscript and how now clarified this in the revised abstract. We agree that collecting both sets of data in the same experiment would be a useful validation of the current results and have acknowledged this in a new Limitations and future directions section of the Discussion of the revised manuscript.

Line 501 - Another limitation of the current study is that the EEG recordings were collected in the separate experiment to the behavioural and pupillometry data. The stimuli and task were similar between experiments, but not identical. For example, the EEG experiment employed coloured arc stimuli presented at a constant rate of ~3.3 Hz and participants were tasked with counting the number of stimuli presented at a target location. By contrast, in the behavioural experiment, participants viewed white blobs presented at an average rate of ~2.8 Hz and performed a binary spatial task coupled with an infrequent reproduction task. An advantage of this was that the sensory responses to stimuli in the EEG recordings were not conflated with motor responses; however, future work combining these measures in the same experiment would serve as a validation for the current results.

(3) How is the arbitrary choice of restricting EEG decoding to a small subset of parieto-occipital electrodes justified? Blinks and other artifacts could have been corrected with proper algorithms (e.g., ICA) (Zhang & Luck, 2025) or even left in, as decoders are not necessarily affected by noise. Moreover, trials with blinks occurring at the stimulus time should be better removed, and the arbitrary selection of a subset of electrodes, while reducing the information in input to the decoder, does not account for trials in which a stimulus was missed (e.g., due to blinks).

Electrode selection was based on several factors: 1) reduction of eye movement/blink artifacts (as noted in the original manuscript), 2) consistency with the previous EEG study (Rideaux, 2024) and other similar decoding studies (Buhmann et al., 2024; Harrison et al., 2023; Rideaux et al., 2023), 3) improved signal-to-noise by including only sensors that carry the most position information (as shown in Supplementary Figure 1a and the previous EEG study). We agree that this was insufficiently explained in the original manuscript and have clarified our sensor selection in the revised version.

Line 631 - We only included the parietal, parietal-occipital, and occipital sensors in the analyses to i) reduce the influence of signals produced by eye movements, blinks, and non-sensory cortices, ii) for consistency with similar previous decoding studies (Buhmann et al., 2024; Rideaux, 2024; Rideaux et al., 2025), and iii) to improve decoding accuracy by restricting sensors to those that carried spatial position information (Supplementary Fig. 1a).

(4) The artifact that appears in many of the decoding results is puzzling, and I'm not fully convinced by the speculative explanation involving slow fluctuations. I wonder if a different high-pass filter (e.g., 1 Hz) might have helped. In general, the nature of this artifact requires better clarification and disambiguation.

We agree that the nature of this artifact requires more clarification and disambiguation. Due to relatively slow changes in the neural signal, which are not stimulus-related, there is a degree of temporal autocorrelation in the recordings. This can be filtered out, for example, by using a stricter high-pass filter; however, we tried a range of filters and found that a cut-off of at least 0.7 Hz is required to remove the artifact, and even a filter of 0.2 Hz introduces other (stimulus-related) artifacts, such as above-chance decoding prior to stimulus onset. These stimulus-related artifacts are due to the temporal smearing of data, introduced by the filtering, and have a more pronounced and complex influence on the results and are more difficult to remove through other means, such as the baseline correction applied in the original manuscript.

The temporal autocorrelation is detected by the decoder during training and biases it to classify/decode targets that are presented nearby in time as similar. That is, it learns the neural pattern for a particular stimulus location based on the activity produced by the stimulus and the temporal autocorrelation (determined by slow stimulus unrelated fluctuations). The latter only accounts for a relatively smaller proportion of the variance in

the neural recordings under normal circumstances and would typically go undetected when simply plotting decoding accuracy as a function of position. However, it becomes weakly visible when decoding accuracy is plotted as a function of distance from the previous target, as now the bias (towards temporally adjacent targets) aligns with the abscissa. Further, it becomes highly visible when the stimulus labels are shuffled, as now the decoder can only learn from the variance associated with the temporal autocorrelation (and not from the activity produced by the stimulus).

In the linear discriminant analysis, this led to temporally proximal items being more likely to be classified as on the same side. This is why there is above-chance performance for repeat trials (Supplementary Figure 2b), and below-chance performance for alternate trials, even when the labels are shuffled – the temporal autocorrelation produces a general bias towards classifying temporally proximate stimuli as on the same side, which selectively improves the classification accuracy of repeat trials. Fortunately, the bias is relatively constant as a function of time within the epoch and is straightforward to estimate by shuffling the labels, which means that it can be removed through a baseline correction. However, to further demonstrate that the autocorrelation confound cannot account for the differences observed between repeat and alternate trials in the micro classification analysis, we now additionally show the results from a more strictly filtered version of the data (0.7 Hz). These results show a similar pattern as the original, with the additional stimulus-related artifacts introduced by the strict filter, e.g., above chance decoding prior to stimulus onset.

In the inverted encoding analysis, the same temporal autocorrelation manifests as temporally proximal trials being decoded as more similar locations. This is why there is increased decoding accuracy for targets with small angular offsets from the previous target, even when the labels are shuffled (Supplementary Figure 3c), because it is on these trials that the bias happens to align with the correct position. This leads to an attractive bias towards the previous item, which is most prominent when the labels are shuffled.

To demonstrate the phenomenon, we simulated neural recordings from a population of tuning curves and performed the inverted encoding analysis on a clean version of the data and a version in which we introduced temporal autocorrelation. We then repeated this after shuffling the labels. The simulation produced very similar results to those we observed in the empirical data, with a single exception: while precision in the simulated shuffled data was unaffected by autocorrelation, precision in the unshuffled data was clearly affected by this manipulation. This may explain why we did not find a correlation between the shuffled and unshuffled precision in the original manuscript.

These results echo those from the classification analysis, albeit in a more continuous space. However, whereas in the classification analysis it was straightforward to perform a baseline correction to remove the influence of general temporal dependency, the more complex nature of the accuracy, precision, and bias parameters over the range of time and delta location makes this approach less appropriate. For example, the bias in the shuffled condition ranged from -180 to 180 degrees, which when subtracted from the bias in the unshuffled condition would produce an equally spurious outcome, i.e., the equal opposite of this extreme bias. Instead for the inverted encoding analysis, we used the data high-pass filtered at 0.7 Hz. As with the classification analysis, this removed the influence of general temporal dependencies, as indicated by the results of the shuffled data analysis (Supplementary Figure 3f), but it also temporally smeared the stimulus-related signal, resulting in above chance decoding accuracy prior to stimulus onset (Supplementary Figure 3d). However, given that we were primarily interested in the pattern of accuracy, precision, and bias as a function of delta location, and less concerned with the precise temporal dynamics of these changes, which appeared relatively stable in the filtered data. Thus, this was the more suitable approach to removing the general temporal dependencies in the inverted encoding analysis and the one that is presented in Figure 3.

We have updated the revised manuscript in light of these changes, including a fuller description of the artifact and the results from the abovementioned control analyses.

Figure 3 updated.

Figure 3 caption - e) Decoding accuracy for stimulus location, from reanalysis of previously published EEG data (17). Inset shows the EEG sensors included in the analysis (blue dots), and black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). f) Decoding accuracy for location, as a function of time and D location. Bright colours indicate higher decoding accuracy; absolute accuracy values can be inferred from (e). g-i) Average location decoding (g) accuracy, (h) precision, and (h) bias from 50 – 500 ms following stimulus onset. Horizontal bar in (e) indicates cluster corrected periods of significance; note, all time points were significantly above chance due to temporal smear introduced by strict high-pass filtering (see Supplementary Figure 3 for full details). Note, the temporal abscissa is aligned across (e & f). Shaded regions indicate $\pm$ SEM.

Line 218 - To further investigate the influence of serial dependence, we applied inverted encoding modelling to the EEG recordings to decode the angular location of stimuli. We found that decoding accuracy of stimulus location sharply increased from ~60 ms following stimulus onset (Fig. 3e). Note, to reduce the influence of general temporal dependencies, we applied a 0.7 Hz high-pass filter to the data, which temporally smeared the stimulus-related information, resulting in above chance decoding accuracy prior to stimulus presentation (for full details, see Supplementary Figure 3). To understand how serial dependence influences the representation of these features, we inspected decoding accuracy for location as a function of both time and D location (Fig. 3f). We found that decoding accuracy varied depending not only as a function of time, but also as a function of D location. To characterise this relationship, we calculated the average decoding accuracy from 50 ms until the end of the epoch (500 ms), as a function of D location (Fig. 3g). This revealed higher accuracy for targets with larger D location. We found a similar pattern of results for decoding precision (Fig. 3h). These results are consistent with the micro temporal context (behavioural) results, showing that targets that alternated were recalled more precisely. Lastly, we calculated the decoding bias as a function of D location and found a clear repulsive bias away from the previous item (Fig. 3i). While this result is inconsistent with the attractive behavioural bias, it is consistent with recent studies of serial dependence suggesting an initial pattern of repulsion followed by an attractive bias during the response period (20–22).

Line 726 - As shown in Supplementary Figure 3, we found the same general temporal dependencies in the decoding accuracy computed using inverted encoding that were found using linear discriminant classification. However, as a baseline correction would not have been appropriate or effective for the parameters decoded with this approach, we instead used a high-pass filter of 0.7 Hz to remove the confound, while being cautious about interpreting the timing of effects produced by this analysis due to the temporal smear introduced by the filter.

Supplementary Figure 2 updated.

Supplementary Figure 2 caption - Removal of general micro temporal dependencies in EEG responses. We found that there were differences in classification accuracy for repeat and alternate stimuli in the EEG data, even when stimulus labels were shuffled. This is likely due to temporal autocorrelation within the EEG data due to low frequency signal changes that are unrelated to the decoded stimulus dimension. This signal trains the decoder to classify temporally proximal stimuli as the same class, leading to a bias towards repeat classification. For example, in general, the EEG signal during trial one is likely to be more similar to that during trial two than during trial ten, because of low frequency trends in the recordings. If the decoder has been trained to classify the signal associated with trial one as a leftward

stimulus, then it will be more likely to classify trial two as a leftward stimulus too. These autocorrelations are unrelated to stimulus features; thus, to isolate the influence of stimulus-specific temporal context, we subtracted the classification accuracy produced by shuffling the stimulus labels from the unshuffled accuracy (as presented in Figure 2e, f). We confirmed that using a stricter high-pass filter (0.7 Hz) removes this artifact, as indicated by the equal decoding accuracy between the two shuffled conditions. However, the stricter high-pass filter temporally smears the stimulus-related signal, which introduces other (stimulus-related) artifacts, e.g., above-chance decoding accuracy prior to stimulus presentation, that are larger and more complex, i.e., changing over time. Thus, we opted to use the original high pass filter (0.1 Hz) and apply a baseline correction. a) The uncorrected classification accuracy along task related and unrelated planes. Note that these results are the same as the corrected version shown in Figure 2e, because the confound is only apparent when accuracy is grouped according to temporal context.

b) Same as (a), but split into repeat and alternate stimuli, along (left) task-related and (right) unrelated planes. Classification accuracy when labels are shuffled is also shown. Inset in (a) shows the EEG sensors included in the analysis (blue dots). (c, d) Same as (a, b), but on data filtered using a 0.7 Hz high-pass filter. Black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). Shaded regions indicate $\pm$ SEM.

Supplementary Figure 3 updated.

Supplementary Figure 3 caption - Removal of general temporal dependencies in EEG responses for inverted encoding analyses. As described in Methods - Neural Decoding, we used inverted encoding modelling of EEG recordings to estimate the decoding accuracy, precision, and bias of stimulus location. Just as in the linear discriminant classification analysis, we also found the influence of general temporal dependencies in the results produced by the inverted encoding analysis. In particular, there was increased decoding accuracy for targets with low D location. This was weakly evident in the period prior to stimulus presentation, but clearly visible when the labels were shuffled. These results are mirror those from the classification analysis, albeit in a more continuous space. However, whereas in the classification analysis it was straightforward to perform a baseline correction to remove the influence of general temporal dependency, the more complex nature of the accuracy, precision, and bias parameters over the range of time and D location makes this approach less appropriate. For example, the bias in the shuffled condition ranged from -180° to 180° , which when subtracted from the bias in the unshuffled condition would produce an equally spurious outcome, i.e., the equal opposite of this extreme bias. Instead for the inverted encoding analysis, we used the data high-pass filtered at 0.7 Hz. As with the classification analysis, this significantly reduced the influence of general temporal dependencies, as indicated by the results of the shuffled data analysis, but it also temporally smeared the stimulus-related signal, resulting in above chance decoding accuracy prior to stimulus onset. However, we were primarily interested in the pattern of accuracy, precision, and bias as a function of D location, and less concerned with the precise temporal dynamics of these changes. Thus, this was the more suitable approach to removing the general temporal dependencies in the inverted encoding analysis and the one that is presented in Figure 3. (a) Decoding accuracy as a function of time for the EEG data filtered using a 0.1 Hz high-pass filter. Inset shows the EEG sensors included in the analysis (blue dots), and black rectangles indicate the timing of stimulus presentations (solid: target stimulus, dashed: previous and subsequent stimuli). (b, c) The same as (a), but as a function of time and D location for (b) the original data and (c) data with shuffled labels. (d-f) Same as (a-c), but for data filtered using a 0.7 Hz high-pass filter. Shaded regions in (a, d) indicate $\pm$ SEM. Horizontal bars in (a, d) indicate cluster corrected periods of significance; note, all time points in (d) were significantly above chance. Note, the temporal abscissa is vertically aligned across plots (a-c & d-f).

In the process of performing these additional analyses and simulations, we became aware that the sign of the decoding bias in the inverted encoding analyses had been interpreted in the wrong direction. That is, where we previously reported an initial attractive bias followed by a repulsive bias relative to the previous target, we have in fact found the opposite, an initial repulsive bias followed by an attractive bias relative to the previous target. Based on the new control analyses and simulations, we think that the latter attractive bias was due to general temporal dependencies. That is, in the filtered data, we only observe a repulsive bias. While the bias associated with serial dependence was not a primary feature of the study, this (somewhat embarrassing) discovery has led to reinterpretation of some results relating to serial dependence. However, it is encouraging to see that our results now align with those of recent studies (Fischer et al., 2024; Luo et al., 2025; Sheehan et al. 2024).

Line 385 - Our corresponding EEG analyses revealed better decoding accuracy and precision for stimuli preceded by those that were different and a bias away from the previous stimulus. These results are consistent with finding that alternating stimuli are recalled more precisely. Further, while the repulsive pattern of biases is inconsistent with the observed behavioural attractive biases, it is consistent with recent work on serial dependence indicating an initial period of repulsion, followed by an attractive bias during the response period (20–22). These findings indicate that serial dependence and first-order sequential dependencies can be explained by the same underlying principle.

(5) Given the relatively early decoding results and surprisingly early differences in decoding peaks, it would be useful to visualize ERPs across conditions to better understand the latencies and ERP components involved in the task.

A rapid presentation design was used in the EEG experiment, and while this is well suited to decoding analyses, unfortunately we cannot resolve ERPs because the univariate signal is dominated by an oscillation at the stimulus presentation frequency (~3 Hz). We agree that this could be useful to examine in future work.

(6) It is unclear why the precision derived from IEM results is considered reliable while the accuracy is dismissed due to the artifact, given that both seem to be computed from the same set of decoding error angles (equations 8-9).

This point has been addressed in our response to point (4).

(7) What is the rationale for selecting five past events as the meso-scale? Prior history effects have been shown to extend much further back in time (Fritzsche et al., 2020).

We used five previous items in the meso analyses to be consistent with previous research on sequential dependencies (Bertelson, 1961; Gao et al., 2009; Jentzsch & Sommer, 2002; Kirby, 1976; Remington, 1969). However, we agree that these effects likely extend further and have acknowledged this in the revised version of the manuscript.

Line 240 - Higher-order sequential dependencies are an example of how stimuli (at least) as far back as five events in the past can shape the speed and task accuracy of responses to the current stimulus (9, 10); however, note that these effects have been observed for more than five events (20).

(8) The decoding bias results, particularly the sequence of attraction and repulsion, appear to run counter to the temporal dynamics reported in recent studies (Fischer et al., 2024; Luo et al., 2025; Sheehan & Serences, 2022).

This point has been addressed in our response to point (4).

(9) *The repulsive component in the decoding results (e.g., Figure 3h) seems implausibly large, with orientation differences exceeding what is typically observed in behavior.*

As noted in our response to point (4), this bias was likely due to the general temporal dependency confound and has been removed in the revised version of the manuscript.

(10) *The pattern of accuracy, response times, and precision reported in Figure 3 (also line 188) resembles results reported in earlier work (Stewart, 2007) and in recent studies suggesting that integration may lead to interference at intermediate stimulus differences rather than improvement for similar stimuli (Ozkirli et al., 2025).*

Thank you for bringing this to our attention, we have acknowledged this in the revised manuscript.

Line 197 - Consistent with our previous binary analysis, and with previous work (19), we also found that responses were faster and more accurate when D location was small (Fig. 3b, c).

(11) *Some figures show larger group-level variability in specific conditions but not others (e.g., Figures 2b-c and 5b-c). I suggest reporting effect sizes for all statistical tests to provide a clearer sense of the strength of the observed effects.*

Yes, as noted in the original manuscript, we find significant differences between the variance task-related and -unrelated conditions. We think this is due to opposing forces in the task-related condition:

“The increased variability of response time differences across the taskrelated plane likely reflects individual differences in attention and prioritization of responding either quickly or accurately. On each trial, the correct response (e.g., left or right) was equally probable. So, to perform the task accurately, participants were motivated to respond without bias, i.e., without being influenced by the previous stimulus. We would expect this to reduce the difference in response time for repeat and alternate stimuli across the taskrelated plane, but not the task-unrelated plane. However, attention may amplify the bias towards making faster responses for repeat stimuli, by increasing awareness of the identity of stimuli as either repeats or alternations (17). These two opposing forces vary with task engagement and strategy and thus would be expected produce increased variability across the task-related plane.” We agree that providing effect sizes may provided a clearer sense of the observed effects and have done so in the revised version of the manuscript.

Line 739 - For Wilcoxon signed rank tests, the rank-biserial correlation (r) was calculated as an estimate of effect size, where 0.1, 0.3, and 0.5 indicate small, medium, and large effects, respectively (54). For Friedman’s ANOVA tests, Kendal’s W was calculated as an estimate of effect size, where 0.1, 0.3, and 0.5 indicate small, medium, and large effects, respectively (55).

(12) *The statement that “serial dependence is associated with sensory stimuli being perceived as more similar” appears inconsistent with much of the literature suggesting that these effects occur at post-perceptual stages (Barbosa et al., 2020; Bliss et al., 2017; Ceylan et al., 2021; Fischer et al., 2024; Fritsche et al., 2017; Sheehan & Serences, 2022).*

In light of the revised analyses, this statement has been removed from the manuscript.

(13) *If I understand correctly, the reproduction bias (i.e., serial dependence) is estimated on a small subset of the data (10%). Were the data analyzed by pooling across subjects?*

The dual reproduction task only occurred on 10% of trials. There were approximately 2000 trials, so ~200 reproduction responses. For the micro and macro analyses, this was sufficient to estimate precision within each of the experimental conditions (repeat/alternate, expected/unexpected). However, it is likely that we were not able to reproduce the effect of precision at the meso level across both experiments because we lacked sufficient responses to reliably estimate precision when split across the eight sequence conditions. Despite this, the data was always analysed within subjects.

(14) I'm also not convinced that biases observed in forced-choice and reproduction tasks should be interpreted as arising from the same process or mechanism. Some of the effects described here could instead be consistent with classic priming.

We agree that the results associated with the forced-choice task (response time task accuracy) were likely due to motor priming, but that a separate (predictive) mechanism may explain the (precision) results associated with the reproduction task. These are two mechanisms we think are operating across the three temporal scales investigated in the current study.

Reviewing Editor Comments:

(1) Clarify task design and measurement: The dense presentation makes it difficult to understand key design elements and their implications. Please provide clearer descriptions of all task elements, and how they relate to each other (EEG vs. behaviour, stimulus plane vs. TR and TU plane, reproduction vs. discrimination and role of priming), and clearly explain how key measures were computed for each of these (e.g., precision, accuracy, reproduction bias).

In the revised manuscript, we have expanded on descriptions of the source and nature of the data (behavioural and EEG), the different planes analyzed in the behavioural task, and how key metrics (e.g., precision) were computed.

(2) Offer more insight into underlying data, including original ERP waveforms to aid interpretation of decoding results and the timing of effects. In particular, unpack the decoding temporal confound further.

In the revised manuscript, we have considerably offered more insight into the decoding results, in particular, the nature of the temporal confound. We were unable to assess ERPs due to the rapid presentation design employed in the EEG experiment.

(3) Justify arbitrary choices such as electrode selection for EEG decoding (e.g., limiting to parieto-occipital sensors), number of trials in meso scale, and the time terminology itself.

In the revised manuscript, we have clarified the reasons for electrode selection.

(3) Discuss deviations from literature: Several findings appear to contradict or diverge from previous literature (e.g., effects of serial dependence). These discrepancies could be discussed in more depth.

Upon re-analysis of the serial dependence bias and removal of the temporal confound, the results of the revised manuscript now align with those from previous literature, which has been acknowledged.

Reviewer #1 (Recommendations for the authors):

(1) would like to use my reviewer's prerogative to mention a couple of relevant publications.

Galluzzi et al (Journal of Vision, 2022) "Visual priming and serial dependence are mediated by separate mechanisms" suggests exactly that, which is relevant to this study.

Xie et al. (Communications Psychology, 2025) "Recent, but not long-term, priors induce behavioral oscillations in peri-saccadic vision" also seems relevant to the issue of different mechanisms.

Thank you for bringing these studies to our attention. We agree that they are both relevant have referenced both appropriately in the revised version of the manuscript.

Reviewer #2 (Recommendations for the authors):

(1) I find the discussion on attention and awareness (from line 127 onward) somewhat vague and requiring clarification.

We agree that this statement was vague and referred to “awareness” without operationation. We have revised this statement to improve clarity.

Line 135 - However, task-relatedness may amplify the bias towards making faster responses for repeat stimuli, by increasing attention to the identity of stimuli as either repeats or alternations (17).

(2) Line 140: It's hard to argue that there are expectations that the image of an object on the retina is likely to stay the same, since retinal input is always changing.

We agree that retinal input is often changing, e.g., due to saccades, self-motion, and world motion. However, for a prediction to be useful, e.g., to reduce metabolic expenditure or speed up responses, it must be somewhat precise, so a prediction that retinal input will change is not necessarily useful, unless it can specify what it will change to. Given retinal input of x at time t , the range of possible values of x at time $t+1$ (predicting change) is infinite. By contrast, if we predict that $x=x$ at time $t+1$ (no change), then we can make a precise prediction. There is, of course, other information that could be used to reduce the parameter space of predicted change from x at time t , e.g., the value of x at time $t-1$, and we think this drives predictions too. However, across the infinite distribution of changes from x , zero change will occur more frequently than any other value, so we think it's reasonable to assert that the brain may be sensitive to this pattern.

(3) Line 564: The gambler's fallacy usually involves sequences longer than just one event.

Yes, we agree that this phenomenon is associated with longer sequences. This section of the manuscript was in regards to previous findings that were not directly relevant to the current study and has been removed in the revised version.

(4) In the shared PDF, the light and dark cyan colors used do not appear clearly distinguishable.

I expect this is due to poor document processing or low-quality image embeddings. I will check that they are distinguishable in the final version.

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